STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

WATER RESOURCE ENGINEERING

Course Code: 2415305 (S2415305)

Diploma in Civil Engineering • Semester III

Student's Name	
Enrollment No.	
Roll No.	
Branch	Civil Engineering
Semester	III
Session	20____ – 20____
Name of Institute	

Department of Civil Engineering

(Name & address of the institute)

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- **Apparatus** — naam aur specification (capacity, IS code).
- **Theory / Formula** — principle aur formula jo calculation mein use hua.
- **Procedure** — step-wise, past tense mein, chhote vakya.
- **Observation table** — readings, units ke saath; IS code ka reference.
- **Calculation** — formula lagakar steps dikhayein.
- **Result** — answer with unit aur IS permissible limit ke saath.
- **Precautions** — 3 se 4 points.

General instructions

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- Diagram pencil se, scale se banayein; likhai blue/black pen se.
- Index page mein page number bharna na bhoolein — file check ke waqt yahi dekha jata hai.
- Practical exam (ELA) mein viva hota hai — aim, formula aur result yaad rakhein.

Subject specific note

Is subject mein lab/practical paper nahi hota — total 150 marks (100 theory + 50 term work) hote hain, isliye assignment, micro project, seminar aur visit report hi jama karne hote hain.

Index

Is subject ka koi laboratory/practical nahi hai — sirf term work (assignment, micro project, seminar aur visit report) 50 marks ka jama hota hai. Page number index mein bharein.

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02	Assignment 2 — Crop water requirement: duty, delta and consumptive use	5		
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TERM WORK NO. 01

Assignment 1 — Hydrology: average rainfall and runoff

Q1. Define hydrology and explain the hydrological cycle with a neat sketch.

Hydrology is the science that deals with the occurrence, movement, distribution and circulation of water on the earth, including precipitation, evaporation, infiltration, runoff and storage. The hydrological cycle is the continuous circulation of water from the ocean to the atmosphere as vapour, from the atmosphere to the land as precipitation, and back to the ocean as surface runoff and groundwater flow.

The main components of the cycle are: precipitation (rainfall, snow), interception, evaporation and transpiration, infiltration, detention storage, surface runoff and groundwater flow. The water balance equation for a catchment is: $P = R + E + \Delta S$, where P is precipitation, R is runoff, E is evapotranspiration and ΔS is the change in storage.

Q2. The rainfall recorded at four rain gauge stations in a catchment is 12, 18, 22 and 28 mm. Calculate the average rainfall by the arithmetic mean method.

Arithmetic mean method: $P_{avg} = (P■ + P■ + P■ + P■)/n = (12 + 18 + 22 + 28)/4 =$ **20 mm**.

This method is used only when the gauges are evenly distributed and the terrain is plain.

Q3. The rainfall at three stations with Thiessen weights 0.25, 0.35 and 0.40 is 20 mm, 26 mm and 32 mm. Find the average rainfall by the Thiessen polygon method.

$P_{avg} = \Sigma (P_i \times A_i)/\Sigma A_i = (20 \times 0.25) + (26 \times 0.35) + (32 \times 0.40) = 5.00 + 9.10 + 12.80 =$ **26.90 mm**.

The Thiessen polygon method gives better results than the arithmetic mean when the gauges are unevenly distributed, because each gauge is weighted according to its area of influence.

Q4. A catchment of 150 km² receives 900 mm of rainfall in a year. If the annual runoff is 0.42 Mm³, calculate the runoff coefficient.

Runoff coefficient, C = Runoff / Rainfall = (0.42 × 10■ m³) / (900 × 10■³ m × 150 × 10■ m²) = 0.42 × 10■ / 135 × 10■ = **0.31**.

The coefficient is about 0.31 (31%), which is typical of a catchment with moderate vegetation and soil permeability.

Q5. Name the factors affecting runoff and state two methods of computing flood discharge.

Runoff depends on rainfall intensity and duration, catchment area, shape and slope, soil type, land use and vegetation, antecedent moisture condition and evaporation. Flood discharge is estimated by the rational method ($Q = C \cdot i \cdot A$) and by empirical formulae such as Ryve's, Inglis' or Dickens' formula.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 02

Assignment 2 — Crop water requirement: duty, delta and consumptive use

Q1. Define duty and delta and derive the relation between them.

Duty (D) is the area of land in hectares irrigated by a continuous flow of 1 m³/s (1 cumec) for the whole base period, expressed in ha/cumec. Delta (Δ) is the total depth of water required by a crop during its base period, expressed in metres.

For a base period B days, the volume of water supplied = 1 × B × 86400 m³, and this water covers D hectares to a depth Δ, so D × Δ × 10■ = 86400 B. Hence Δ = 8.64 B / D (m), or D = 8.64 B / Δ (ha/cumec).

Q2. A crop has a base period of 120 days and a delta of 0.60 m. Find its duty.

D = 8.64 B / Δ = (8.64 × 120) / 0.60 = 1036.8/0.60 = **1728 ha/cumec**.

Q3. Water is supplied at 2 cumecs to a field of 400 hectares for 100 days. Calculate the delta.

Volume supplied = 2 × 100 × 86400 = 17.28 × 10■ m³.

Δ = Volume / Area = 17.28 × 10■ / (400 × 10■) = **4.32 m**.

Duty, D = 8.64 × 100 / 4.32 = **200 ha/cumec**.

Q4. Explain consumptive use and the factors affecting it.

Consumptive use (evapotranspiration) is the total water used by a crop for its growth, including transpiration and evaporation from the soil and plant surfaces. It depends on climate (temperature, humidity, wind, sunshine), crop type and stage of growth, soil moisture and irrigation practices. It is estimated by the Blaney–Criddle method, Penman method or by lysimeter measurements.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 03

Assignment 3 — Dams and spillways

Q1. Classify dams and state the factors governing the choice of a dam site.

Dams are classified as: (i) by material — earthen, rockfill, masonry, concrete and composite dams; (ii) by purpose — storage, detention, diversion and check dams; (iii) by structural behaviour — gravity, arch, buttress and embankment dams.

A dam site is selected on the basis of a narrow valley with a wide reservoir basin, sound rock or impervious foundation, availability of construction material, favourable geology, low seismic risk, minimum submergence of land, and accessibility.

Q2. Draw a labelled sketch of a gravity dam section and show the forces acting on it.

The forces acting on a gravity dam are: (i) the weight of the dam, (ii) water pressure on the upstream face, (iii) uplift pressure at the base, (iv) silt pressure, (v) wave pressure, (vi) earthquake forces, and (vii) ice pressure (in cold regions).

A gravity dam resists these forces by its own weight and must satisfy the requirements of stability against overturning, sliding, crushing and tension at the heel.

Q3. Why is a spillway necessary? Name the common types.

A spillway is a safety structure which passes the surplus flood water from the reservoir to the downstream side without damaging the dam. Without a spillway, the water level would rise above the dam crest and endanger the structure. Common types are the ogee (overflow) spillway, chute spillway, side channel spillway, shaft spillway, siphon spillway and emergency spillway.

Q4. Write short notes on the causes and control of seepage through an earthen dam.

Seepage through an earthen dam occurs through the body of the dam, the foundation and the abutments. It may cause erosion (piping), sloughing and reduction of stability. Seepage is controlled by a central clay core, upstream impervious blanket, cut-off trench, toe drain, horizontal drainage blanket, rock toe and grouting of the foundation.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 04

Assignment 4 — Minor and micro irrigation

Q1. Differentiate between flow irrigation and lift irrigation.

In flow irrigation, water is supplied to the fields by gravity from a canal or a river at a higher level, and no pumping is required. In lift irrigation, water is lifted from a river, well or canal by pumps and then distributed to the fields; it is used where the source is lower than the commanded area.

Q2. Explain drip and sprinkler irrigation with their advantages.

Drip (trickle) irrigation applies water drop by drop near the root zone of the plant through emitters and lateral pipes. Sprinkler irrigation sprays water over the crop like rain through nozzles mounted on pipes.

Advantages: water saving of 30–60%, uniform application, reduced weed growth, no land levelling, suitable for undulating land, fertigation possible, and higher crop yield.

Q3. A drip irrigation system is to be designed for 1 hectare with 4000 plants spaced 2.5 m × 1 m. If each plant needs 8 litres per day through 4 emitters of 4 l/h, find the daily operating time.

Discharge per plant = 4 emitters × 4 l/h = 16 l/h.

Time required per plant = 8 l / 16 l/h = 0.5 h = **30 minutes** per day.

Total water requirement = 4000 × 8 = 32,000 l/day = **32 m³/day**.

If the system is operated in two shifts, the capacity of the pump = 32/2 = **16 m³/h**.

Q4. Write a short note on rainwater harvesting.

Rainwater harvesting is the collection and storage of rainwater from roofs and other surfaces for reuse, or its artificial recharge into the ground. For a roof area of 100 m² with 800 mm annual rainfall and a runoff coefficient of 0.85, the harvestable water = 100 × 0.8 × 0.85 = **68 m³ per year**. A first-flush arrangement, filter and storage tank or recharge pit are the main components of the system.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 05

Assignment 5 — Diversion headworks and cross drainage works

Q1. What is a diversion headwork? Draw a labelled sketch and name its components.

A diversion headwork is constructed across a river to raise its water level and to divert the required quantity of water into an off-taking canal. Its main components are the weir or barrage (crest, body wall, downstream floor, cut-off walls), under-sluices, canal head regulator, river training works (guide banks, spurs) and fish ladder.

Q2. Differentiate between a weir and a barrage.

A weir is a solid masonry/concrete wall over which water flows, and its crest level is fixed. A barrage has a series of gates between piers, so that the water level and discharge can be regulated; it provides better control of silt and floods and is preferred on alluvial rivers.

Q3. What are cross drainage works? Name the four main types with their suitability.

Cross drainage works are structures constructed where a canal crosses a natural drainage or river. The four main types are: (i) aqueduct — canal over the drainage; (ii) super-passage — drainage over the canal; (iii) level crossing — both at the same level, with a regulator on each; and (iv) inlet and outlet — the drainage water is admitted into the canal and taken out again downstream.

Q4. Sketch a typical cross section of a canal showing bed width, side slopes, berms, banks and the freeboard.

A canal section consists of the bed, side slopes (usually 1:1 to 1.5:1 in cutting and 1.5:1 in filling), berms, service roads on the banks, and a freeboard (about 0.5–0.75 m) above the full supply level to prevent overtopping.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 06

Micro project — design of a rainwater harvesting system for a college building

1. Objective

To design a rainwater harvesting and artificial recharge system for the roof of the civil engineering block of the institute, so that a part of the water demand is met and the groundwater table is recharged.

2. Data collected

Particular	Value
Roof area (RCC slab)	620 m²
Average annual rainfall of the district	1050 mm
Runoff coefficient for RCC roof	0.85
Number of rainy days	45
Water demand of the building	8,000 litres/day

Rainfall data collected from the district agriculture office / IMD website.

3. Design calculations

Annual harvestable volume = roof area × rainfall × runoff coefficient = 620 × 1.05 × 0.85 = **553.35 m³/year** ≈ **5.5 lakh litres/year**.

Average daily availability over the year = 553,350/365 = **1516 litres/day**. If water is stored only for the rainy season, the storage tank may be designed for 20,000 litres, and the excess water is taken to a recharge pit.

Size of the first-flush tank = 1 mm of rainfall over the roof = 620 × 0.001 = **0.62 m³**.

Recharge pit: for a percolation rate of 0.15 m/h and a peak rainfall intensity of 50 mm/h, the peak flow = 620 × 0.050 × 0.85/3600 = 0.0073 m³/s = 7.3 l/s; a pit of 2 m × 2 m × 3 m filled with pebbles and sand is adequate.

4. Components of the system

- Roof gutters and down-take pipes of 110 mm PVC.
- First-flush valve and wire-mesh filter.
- Storage tank of 20,000 litres capacity with a settling chamber.
- Desilting chamber and recharge pit with a bore for excess water.
- Overflow pipe connected to the campus storm-water drain.

5. Conclusion

The proposed system can harvest about 5.5 lakh litres of water per year, which is sufficient to meet roughly 19% of the annual water demand of the building, and it will also recharge the groundwater. The payback period is about 4 to 5 years considering the prevailing water charges.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 07

Seminar — Applications of remote sensing and GIS in water resource engineering

1. Introduction

Remote sensing is the technique of collecting information about the earth's surface from satellites or aircraft without physical contact, while a Geographic Information System (GIS) is a computer-based tool for storing, analysing and displaying spatial data. Together they provide fast, accurate and repeatable information for planning and managing water resources.

2. Applications

- Mapping of catchment areas, drainage networks and watershed boundaries.
- Estimation of rainfall, snow cover, evaporation and soil moisture from satellite data.
- Identification of suitable sites for dams, reservoirs and recharge structures.
- Flood mapping and flood damage assessment using multi-date satellite images.
- Monitoring of reservoir water spread area and sedimentation.
- Groundwater potential mapping and selection of well sites.
- Canal alignment, command area mapping and irrigation scheduling.
- Land use / land cover change detection for hydrological modelling.

3. Merits and limitations

Merits: large area coverage, repeatability, less field work, digital data suitable for GIS analysis, and quick updating. Limitations: high initial cost, need for trained manpower, cloud cover affecting optical data, and the need for ground-truth verification.

4. Conclusion

Remote sensing and GIS have become essential tools for water resource planning, from watershed development to flood management, and their use should be included in the planning of every irrigation and water supply project.

Date: _____ Teacher's Signature: _____

TERM WORK NO. 08

Visit report — irrigation project / dam site visit

1. Particulars of the visit

Item	Details
Name of the project	_____
Location / district	_____
Date of visit	_____
Type of structure	Dam / weir / barrage / canal head regulator
Officer who explained	_____
Number of students	_____

Visit report ko apne actual visit ke anusaar bharein; yahan diya format reference ke liye hai.

2. Technical details observed

- Catchment area, reservoir capacity and water spread area.
- Type and height of the dam, length of the spillway and number of gates.
- Full reservoir level, maximum water level and dead storage level.
- Canals off-taking from the headworks, their discharge and command area.
- Instruments installed for measurement of water level and discharge.

3. Observations and comments

The condition of the concrete surfaces, gates, hoists and spillway was observed. Seepage markings, vegetation on the earthen portion and the condition of the approach roads were noted. The maintenance work being carried out at the site was also observed.

4. Learning outcome

The visit gave a practical understanding of the working of a diversion headwork and the importance of regular inspection and maintenance of hydraulic structures. Photographs and sketches may be attached with this report.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	S.K. Garg, *Irrigation Engineering and Hydraulic Structures*, Khanna Publishers.
2	P.N. Modi, *Irrigation, Water Resources and Water Power Engineering*, Standard Book House.
3	IS: 6512 — gravity dams; IS: 8826 — spillways; IS: 7114 — canal linings.
4	IS: 10824 — sprinkler irrigation; IS: 10799 — drip irrigation; IS: 15797 — rainwater harvesting.
5	SBTE Bihar curriculum for Water Resource Engineering (2415305 / S2415305).

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.

STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

CONCRETE TECHNOLOGY

Course Code: 2415302 (P2415302)

Diploma in Civil Engineering • Semester III

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Index

Ye file SBTE Bihar curriculum ke P2415302 ke 18 practicals par aadharit hai. Page number index mein bharein aur har practical par date aur signature lein.

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02	Determination of standard consistency of cement using Vicat apparatus (IS: 4031 Part 4)	5		
03	Determination of initial and final setting time of cement using Vicat apparatus (IS: 4031 Part 5)	6		
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PRACTICAL NO. 01

Determination of fineness of cement by sieving (IS: 4031 Part 1)

Aim

To determine the fineness of the given sample of cement by sieving on a 90 µm IS sieve.

Apparatus / Equipment Required

90 µm IS sieve with pan and lid; Weighing balance (accuracy 0.1 g); Sieve shaker / brush, airtight container, sample of cement.

Theory

Fineness of cement is a measure of the size of its particles and affects the rate of hydration, workability, rate of strength gain, shrinkage and heat of hydration. Finer cement hydrates faster and gives higher early strength, but needs more water and increases shrinkage.

In the sieving method a known mass of cement is sieved continuously on a 90 µm IS sieve and the mass of the residue is expressed as a percentage of the original mass. A residue of about 5–10% indicates cement of normal fineness.

Formulae Used

- Percentage residue = (mass of residue / mass of sample) × 100
- Fineness is reported as the percentage of cement retained on the 90 µm sieve.

Procedure

1. The 90 µm sieve was cleaned, dried and weighed along with the pan.
2. A cement sample of 100 g was taken and placed on the sieve; the lid was fixed.
3. The sieve was shaken continuously by hand for 15 minutes, or in a sieve shaker for 10 minutes, with occasional tapping of the sides.
4. The residue left on the sieve was weighed accurately on a balance of 0.1 g accuracy.
5. The operation was repeated for a second sample and the mean percentage residue was calculated.

Observation

Trial	Mass of sample (g)	Mass of residue (g)	Residue (%)
1	100.0	8.4	8.4
2	100.0	8.7	8.7
3	100.0	8.5	8.5
Mean	100.0	8.53	8.53

Cement used: OPC 43 grade. Readings are sample values.

Calculation

Percentage residue = (8.53 / 100.0) × 100 = **8.53 %**

Fineness of the cement in terms of specific surface (Blaine's apparatus) is normally 280–350 m²/kg for OPC 43 grade; the sieving method gives the residue on the 90 µm sieve only.

Result

The fineness of the given cement sample is **8.53 % residue on the 90 µm IS sieve**, which is less than 10% and hence the cement is of satisfactory fineness as per IS: 4031 (Part 1).

Precautions

- The sieve should be perfectly clean and dry before sieving.
- Continuous shaking is necessary; the cement should not be pressed or rubbed by hand.
- Weighing must be done immediately to avoid absorption of moisture from the air.
- Take two samples and report the mean value.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 02

Determination of standard consistency of cement using Vicat apparatus (IS: 4031 Part 4)

Aim

To determine the standard consistency of the given cement paste, i.e. the quantity of water required to produce a paste that allows the Vicat plunger to penetrate 5–7 mm from the bottom.

Apparatus / Equipment Required

Vicat apparatus with plunger (10 mm diameter, 50 mm long); Balance (0.1 g), measuring cylinder (100 ml), non-porous mixing tray, trowel; Stopwatch, 400 g cement sample.

Theory

Standard consistency is the water content of a cement paste, expressed as a percentage of the mass of cement, which will permit the Vicat plunger to penetrate to a point 5–7 mm above the bottom of the mould. It is used to fix the water content for the setting-time and soundness tests.

The water requirement for standard consistency of ordinary Portland cement is usually between 25% and 35% of the mass of cement.

Formulae Used

- Standard consistency, P = (mass of water added / mass of cement) × 100

Procedure

1. 400 g of cement was weighed and placed in the mixing tray.
2. Water was added in stages (starting with 25%) and the paste was mixed thoroughly for 3–5 minutes.
3. The paste was filled in the Vicat mould, levelled and placed on the base plate of the apparatus.
4. The plunger was lowered gently and allowed to penetrate the paste for 30 seconds; the depth of penetration from the bottom was noted with the scale.
5. The trial was repeated with different water contents (28%, 30%, 32%) until the penetration was 5–7 mm.
6. The percentage of water giving the required penetration was recorded as the standard consistency.

Observation

Trial	Mass of cement (g)	Water added (ml)	Penetration from bottom (mm)	P (%)
1	400	100 (25%)	16	25.0
2	400	112 (28%)	11	28.0
3	400	120 (30%)	6	30.0
4	400	128 (32%)	2	32.0

The required penetration of 5–7 mm was obtained at 30% water content.

Calculation

P = (120 / 400) × 100 = **30.0 %**

Since the penetration at this water content was 6 mm (within 5–7 mm), the standard consistency is 30%.

Result

The standard consistency of the given cement is **30 %**, i.e. 30 parts of water are required for every 100 parts of cement (120 ml for 400 g of cement).

Precautions

- The mixing must be done quickly and uniformly; mixing time should be 3–5 minutes.
- The plunger must be cleaned after every trial and the paste must be freshly made for each trial.
- The plunger should be released gently, without any jerk, and readings taken exactly after 30 seconds.
- Use potable water at room temperature (27 ± 2 °C).

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 03

Determination of initial and final setting time of cement using Vicat apparatus (IS: 4031 Part 5)

Aim

To determine the initial and final setting time of the given cement sample by the Vicat apparatus.

Apparatus / Equipment Required

Vicat apparatus with plunger (10 mm) and needle (1 mm square with 1 mm projecting core); Balance, measuring cylinder, mixing tray, trowel, stopwatch, glass plate.

Theory

Setting is the process of stiffening of cement paste after mixing with water. Initial setting time is the interval between the time of mixing water and the time when the Vicat needle fails to penetrate the paste more than 5 ± 0.5 mm from the bottom of the mould. Final setting time is the interval until the needle makes an impression on the surface but the annular ring fails to cut it.

As per IS: 4031, the initial setting time of OPC shall not be less than 30 minutes and the final setting time shall not exceed 600 minutes.

Formulae Used

- Initial setting time = (time at which penetration is 5 ± 0.5 mm) – (time of adding water)
- Final setting time = (time at which the annular ring fails to make an impression) – (time of adding water)

Procedure

1. A cement paste of standard consistency (30% water) was prepared and filled in the Vicat mould.
2. The mould was placed on the base plate and the 1 mm square needle was lowered gently on the surface.
3. The penetration was observed at regular intervals of 5 minutes until the needle failed to penetrate 5 ± 0.5 mm from the bottom; the time was recorded as the initial setting time.
4. The same specimen was then tested with the annular ring needle until the ring failed to make an impression on the surface; this time was noted.
5. The elapsed time for both observations was calculated and compared with IS limits.

Observation

Time from adding water	Observation with needle	Remark
10 min	Full penetration up to bottom	Paste still plastic
30 min	Penetration 12 mm from bottom	Setting started
40 min	Penetration 7 mm from bottom	—
45 min	Penetration 5 mm from bottom	Initial setting time = 45 min
6 h 00 min	Ring makes impression	—
5 h 50 min	Ring fails to make impression	Final setting time = 350 min

Water used = 30% of mass of cement (standard consistency). Readings are sample values.

Calculation

Initial setting time = **45 minutes**

Final setting time = **5 hours 50 minutes = 350 minutes**

Check: initial setting time 45 min > 30 min (minimum limit) ✓

Final setting time 350 min < 600 min (maximum limit) ✓

Result

The initial setting time of the given cement is **45 minutes** and the final setting time is **350 minutes**. Both values are within the limits specified by IS: 4031 (Part 5), hence the cement is suitable for construction work.

Precautions

- The stopwatch should be started at the instant of adding water to the cement.
- The needle must be cleaned and lowered gently at every observation, and should be kept 10 mm inside the edge of the mould.
- The mould must not be disturbed or tapped during the test.
- Maintain room temperature of 27 ± 2 °C and relative humidity of at least 50%.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 04

Determination of compressive strength of cement (IS: 4031 Part 6)

Aim

To determine the compressive strength of the given cement by testing 70.6 mm cement mortar cubes at 7 days and 28 days.

Apparatus / Equipment Required

70.6 mm cube moulds (steel, ISI marked), vibrating machine; Standard sand (Grade I–III as per IS 650), mixing tray, trowel, 1 kg balance with 1 g accuracy; Compression testing machine (2000 kN), curing tank, stopwatch.

Theory

The compressive strength of cement is determined on mortar cubes made of 1 part cement and 3 parts of standard sand by mass, with water equal to (P/4 + 3) percent of the combined mass of cement and sand, where P is the standard consistency.

The cubes are cured under water at 27 ± 2 °C and tested in a compression testing machine. The compressive strength gives an indication of the class (grade) of the cement.

Formulae Used

- Water content = (P/4 + 3) % of (cement + sand) mass
- Compressive strength = load at failure / area of cube (mm²) = P/A

Procedure

1. 200 g cement and 600 g standard sand were mixed dry on a non-porous mixing surface.
2. Water equal to (P/4 + 3)% of 800 g, i.e. about 96 ml, was added and a uniform mortar was prepared.
3. The mortar was filled in 70.6 mm cube moulds in three layers, compacted and vibrated for 2 minutes.
4. The cubes were kept for 24 hours in a moist room and then demoulded and cured in water.
5. Three cubes were taken out after 7 days and three after 28 days and tested in the compression testing machine at a uniform rate of loading.
6. The compressive strength of each cube and the mean strength were calculated.

Observation

Age of curing	Cube No.	Failure load (kN)	Area (mm²)	Strength (N/mm²)
7 days	1	212	4984	42.5
7 days	2	218	4984	43.7
7 days	3	216	4984	43.3
28 days	1	262	4984	52.6
28 days	2	268	4984	53.8
28 days	3	265	4984	53.2

Cement: OPC 43 grade; mortar 1:3 by mass; water content 12% (P = 36%). Area of 70.6 mm cube = 4984 mm². Readings are sample values.

Calculation

Mean 7-day strength = (42.5 + 43.7 + 43.3)/3 = **43.2 N/mm²**

Mean 28-day strength = (52.6 + 53.8 + 53.2)/3 = **53.2 N/mm²**

Sample calculation: 212 kN / 4984 mm² = 212000/4984 = 42.54 N/mm²

As per IS: 4031 (Part 6), 28-day strength of OPC 43 grade shall not be less than 43 N/mm² — the result satisfies this requirement.

Result

The compressive strength of the given cement is **43.2 N/mm² at 7 days** and **53.2 N/mm² at 28 days**. The cement satisfies the 43 grade requirement of IS: 4031 (Part 6).

Precautions

- Use standard sand and accurate proportions; even small errors change the result considerably.
- Compact the mortar uniformly and at the same time for all cubes.
- Keep the cubes covered with wet cloth before demoulding and cure them in water at 27 ± 2 °C.
- The rate of loading must be uniform and the cube must be tested at right angles to the compacting face.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 05

Determination of soundness of cement by Le-Chatelier method (IS: 4031 Part 3)

Aim

To determine the soundness of the given cement by measuring the expansion of the set cement paste in the Le-Chatelier apparatus.

Apparatus / Equipment Required

Le-Chatelier apparatus (set of 6 moulds with weights and glass plates); Water bath, balance, measuring cylinder, mixing tray, trowel, Vicat apparatus.

Theory

A cement is said to be sound if it does not undergo excessive expansion after setting. Unsoundness is caused by uncombined lime and magnesia, and by excessive gypsum, which expand on hydration and cause cracking of concrete.

In the Le-Chatelier test, a cement paste of standard consistency is placed in the split mould and the distance between the two pointers is measured before and after boiling. The difference gives the expansion, which shall not exceed 10 mm for ordinary cement.

Formulae Used

- Expansion = (distance between pointers after boiling) – (distance before boiling)
- Soundness is satisfactory if expansion ≤ 10 mm.

Procedure

1. The Le-Chatelier mould was assembled, lightly oiled and opened out on a glass plate.
2. A cement paste of standard consistency (30% water) was prepared and filled in the mould.
3. The mould was covered with a glass plate, kept for 24 hours in moist air and then the distance between the pointers was measured.
4. The mould was then immersed in water and heated to boiling in 27 ± 3 minutes and kept boiling for 3 hours.
5. The mould was cooled and the distance between the pointers was measured again.
6. The expansion was calculated and compared with the permissible limit.

Observation

Sample	Pointer distance before (mm)	Pointer distance after (mm)	Expansion (mm)
1	12.5	14.0	1.5
2	12.6	14.1	1.5
Mean	12.55	14.05	1.5

Cement: OPC 43 grade. The initial 24-hour curing was done at 27 ± 2 °C in a moist chamber.

Calculation

Expansion = 14.0 – 12.5 = **1.5 mm** (sample 1); 14.1 – 12.6 = 1.5 mm (sample 2)

Mean expansion = **1.5 mm**

Permissible limit = 10 mm; 1.5 mm < 10 mm, hence the cement is sound.

Result

The expansion of the given cement is **1.5 mm**, which is much less than the permissible limit of 10 mm. Hence the cement is **sound** and does not contain free lime in harmful quantity.

Precautions

- Fill the mould carefully so that the split is not distorted.
- The paste must be at standard consistency, otherwise the expansion value changes.
- Heat the water bath gradually so that boiling is reached in 27 ± 3 minutes.
- Measure the pointer distance with a vernier caliper or scale held exactly at the same position.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 06

Sieve analysis of fine and coarse aggregates — grading and fineness modulus (IS: 2386 Part 1)

Aim

To determine the size, grading and fineness modulus of the given fine and coarse aggregates by sieve analysis.

Apparatus / Equipment Required

Set of IS sieves (4.75 mm to 150 µm for fine and 80 mm to 2.36 mm for coarse aggregate); Sieve shaker, weighing balance (0.1 g), tray, brush, oven.

Theory

Sieve analysis shows the distribution of aggregate particles by size and governs the grading of the aggregate; a well-graded aggregate gives a dense and workable concrete. The fineness modulus (FM) is the sum of the cumulative percentages retained on a standard series of sieves divided by 100, and is an index of fineness — fine sand has FM about 2.0–2.6, medium sand 2.6–2.9 and coarse sand above 2.9.

Formulae Used

- Percentage retained = (mass retained on a sieve / total mass) × 100
- Cumulative percentage retained = sum of the percentages retained on that sieve and all coarser sieves
- Fineness modulus = (Σ cumulative % retained) / 100

Procedure

1. The sample was dried in an oven at 100–110 °C and exactly 1000 g of fine aggregate was weighed.
2. The sieves were placed in a nest in descending order (4.75 mm at top) and the sample was sieved for 10 minutes in a sieve shaker.
3. The mass retained on each sieve was weighed and recorded.
4. The operation was repeated for 5 kg of coarse aggregate with the coarse sieve set.
5. The percentage retained, cumulative percentage retained and fineness modulus were calculated, and the grading was plotted on a semi-log graph.

Observation

IS sieve	Fine agg. retained (g)	Cum. % retained	Coarse agg. retained (g)	Cum. % retained
20 mm	—	—	400	8
12.5 mm	—	—	2350	55
10 mm	—	—	1150	78
4.75 mm	20	2	750	93
2.36 mm	80	10	250	98
1.18 mm	180	28	100	100
600 µm	260	54	—	100
300 µm	250	79	—	100
150 µm	170	96	—	100

Sample masses: fine aggregate 1000 g, coarse aggregate 5000 g. Sieves coarser than 20 mm retained nothing. Readings are sample values.

Calculation

Fine aggregate: Σ cumulative % retained = 2 + 10 + 28 + 54 + 79 + 96 = 269, hence FM = 269/100 = **2.69**; the grading satisfies the Zone II limits of IS 383, so the sand is **Zone II (medium sand)**.

Coarse aggregate: Σ cumulative % retained = 0 + 0 + 8 + 55 + 78 + 93 + 98 + 100 + 100 + 100 + 100 = 732

FM of coarse aggregate = 732/100 = **7.32**; nominal size = 20 mm.

Result

The fineness modulus of the given fine aggregate is **2.69** (Zone II, medium sand) and that of the coarse aggregate is **7.32** for a nominal size of 20 mm. Both aggregates are well graded and suitable for concrete.

Precautions

- The sample must be oven-dried before sieving; wet aggregate gives wrong results.
- Do not overload the sieves — the mass retained should not exceed the recommended limit.
- Shake the sieves for the specified time and brush the underside of each sieve.
- Avoid loss of material during handling; check that the sum of masses equals the sample mass.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 07

Determination of flakiness index and elongation index of coarse aggregate (IS: 2386 Part 1)

Aim

To determine the flakiness index and elongation index of the given coarse aggregate sample.

Apparatus / Equipment Required

Thickness (flakiness) gauge and length (elongation) gauge; Weighing balance (0.1% accuracy), IS sieves 63, 50, 40, 31.5, 25, 20, 16, 12.5, 10 mm and 6.3 mm; Tray, brush, sample of coarse aggregate.

Theory

Flaky particles have a thickness less than 0.6 times their mean dimension, while elongated particles have a length greater than 1.8 times their mean dimension. Such particles reduce workability, strength and durability of concrete because they pack badly and break easily.

The flakiness index is the percentage by mass of particles passing the thickness gauge; the elongation index is the percentage by mass of particles retained on the length gauge.

Formulae Used

- Flakiness index = (mass passing thickness gauge / total mass of sample) × 100
- Elongation index = (mass retained on length gauge / total mass of sample) × 100

Procedure

1. The sample was sieved and the material passing the 6.3 mm sieve and retained on the 63 mm sieve was discarded.
2. About 2000 g of the sized aggregate was taken and weighed.
3. Each particle was tested on the thickness gauge; particles passing through the gauge were separated and weighed.
4. The remaining particles were tested on the length gauge, and particles retained on the gauge were weighed.
5. The flakiness index and elongation index were calculated as percentages of the sample mass.

Observation

Test	Mass of sample (g)	Mass of flaky/elongated particles (g)	Index (%)
Flakiness	2000	360	18.0
Elongation	2000	250	12.5

Aggregate size 20 mm nominal. Readings are sample values.

Calculation

Flakiness index = (360 / 2000) × 100 = **18.0 %**

Elongation index = (250 / 2000) × 100 = **12.5 %**

Combined flakiness and elongation index = 18.0 + 12.5 = **30.5 %**, which is within the limit of 40% generally specified for concrete aggregates.

Result

The flakiness index of the given coarse aggregate is **18.0 %** and the elongation index is **12.5 %**. The combined index (30.5%) is within permissible limits, hence the aggregate is suitable for concrete work.

Precautions

- Test each particle singly and pass it through the gauge without forcing.
- Use the correct gauge for the size group being tested.
- Weigh the separated particles accurately and keep the sample free from dust.
- Take a representative sample by quartering.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 08

Determination of silt content in sand (IS: 2386 Part 2)

Aim

To determine the percentage of silt and clay present in the given sample of sand.

Apparatus / Equipment Required

250 ml measuring cylinder, 1000 ml jar, weighing balance; Common salt solution (1%), glass rod, oven, tray.

Theory

Sand may contain silt and clay which coat the particles and increase the water demand, thereby reducing the strength and durability of concrete. As per IS: 2386 (Part 2), the silt content should not exceed 3% by mass for concrete work (up to 5% may be allowed where the concrete is not subjected to abrasion).

When sand is stirred in salt water, the silt remains in suspension while the sand settles quickly, and the volume of silt can be read from the graduated cylinder.

Formulae Used

- Silt content by volume = (volume of silt layer / volume of sand layer) × 100

Procedure

1. A 250 ml measuring cylinder was filled up to 100 ml with the sand sample.
2. Salt solution was added up to 150 ml and the contents were stirred thoroughly with a glass rod.
3. The cylinder was kept undisturbed for 3 hours so that the silt settled above the sand.
4. The volume of the silt layer and that of the sand layer were read from the graduations.
5. The test was repeated and the mean silt content was calculated.

Observation

Trial	Volume of sand (ml)	Volume of silt (ml)	Silt content (%)
1	100	4.0	4.0
2	100	3.8	3.8
3	100	4.2	4.2
Mean	100	4.0	4.0

Salt solution (1%) is used instead of plain water so that clay particles do not flocculate.

Calculation

Silt content = (4.0 / 100) × 100 = **4.0 % by volume**

Since 4.0% is more than the 3% limit for concrete work, the sand should be washed before use.

Result

The silt content of the given sand sample is **4.0 %** by volume. This is slightly above the permissible limit of 3% for concrete work, hence the sand requires washing before use.

Precautions

- Stir the sample thoroughly so that all silt goes into suspension.
- Do not disturb the cylinder while the silt is settling.
- Read the levels with the eye at the same height as the meniscus.
- Take a representative sample of sand and remove pebbles before testing.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 09

Determination of bulking of sand (IS: 2386 Part 3)

Aim

To determine the percentage of bulking of the given sand at different moisture contents.

Apparatus / Equipment Required

250 ml measuring cylinder, weighing balance, oven, tray; Measuring jar, glass rod, sample of air-dry sand.

Theory

Bulking of sand is the increase in the volume of sand due to the film of water surrounding its particles when it becomes moist. Bulking makes the measuring of sand by volume unreliable and can lead to deficiency of sand in the mix; hence the volume must be corrected.

Bulking is maximum at a moisture content of about 4–6% (up to 20–30% for fine sand) and disappears when the sand is fully saturated or completely dry.

Formulae Used

- Bulking (%) = [(h■ – h■) / h■] × 100
- where h■ = height of sand in the cylinder when dry and h■ = height when moist

Procedure

1. The sand was dried in an oven and 200 ml of it was filled in a measuring cylinder without tamping; the height was recorded.
2. Water was added in increments of 1%, 2%, 4% and 6% by mass and the sand was mixed thoroughly each time.
3. After each addition the sand was poured into the cylinder and the height was measured.
4. The bulking percentage was calculated for each moisture content.
5. A graph of bulking percentage against moisture content was plotted.

Observation

Moisture content (%)	Height dry h■ (ml)	Height moist h■ (ml)	Bulking (%)
0	200	200	0.0
1	200	212	6.0
2	200	228	14.0
4	200	244	22.0
6	200	248	24.0
Saturated	200	204	2.0

Bulking is maximum (24%) at about 6% moisture content and reduces when the sand is saturated.

Calculation

At 4% moisture: bulking = (244 – 200)/200 × 100 = **22.0 %**

At 6% moisture: bulking = (248 – 200)/200 × 100 = **24.0 %**

Correction for volume batching: if 1 m³ of dry sand is required, the volume of moist sand to be measured = 1 × (1 + 0.24) = **1.24 m³** at 6% moisture.

Result

The bulking of the given sand is maximum, **24 %**, at a moisture content of about 6%. For volume batching, the measured volume of moist sand must be increased by 24% to get the same quantity of dry sand.

Precautions

- Fill the cylinder loosely and do not tamp the sand, otherwise bulking is underestimated.
- Mix the water uniformly through the sample and allow it to stand for a few minutes before measuring.
- Use the same cylinder and the same method of filling throughout the test.
- Take readings at the same eye level to avoid parallax error.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 10

Determination of bulk density and voids of fine and coarse aggregates (IS: 2386 Part 3)

Aim

To determine the bulk density (loose and compacted) and percentage of voids of the given fine and coarse aggregates.

Apparatus / Equipment Required

Cylindrical metal measure of 3 litre / 15 litre capacity; Weighing balance (accuracy 0.1%), tamping rod (16 mm diameter, 600 mm long), oven, tray.

Theory

Bulk density is the mass of aggregate required to fill a unit volume, including the voids between particles. It is needed for converting mass proportions into volume proportions and for checking the yield of concrete.

The percentage of voids is the ratio of the volume of voids to the total volume of the aggregate and is calculated from the bulk density and the specific gravity of the aggregate.

Formulae Used

- Bulk density = mass of aggregate filling the measure / volume of measure (kg/m³ or kg/litre)
- Voids (%) = [(specific gravity × 1000 – bulk density) / (specific gravity × 1000)] × 100

Procedure

1. The empty measure was weighed and its volume was determined (3 litres for fine aggregate).
2. The aggregate was filled in the measure up to the top and struck off level with a straight edge; the measure was weighed to obtain the loose bulk density.
3. For compacted bulk density, the aggregate was filled in three layers, each layer being tamped 25 times.
4. The measure was weighed again and the compacted bulk density was calculated.
5. The percentage of voids was computed using the specific gravity of the aggregate.

Observation

Aggregate	Mass of aggregate (kg)	Volume of measure (l)	Bulk density (kg/m³)
Fine — loose	4.62	3	1540
Fine — compacted	4.92	3	1640
Coarse — loose	23.10	15	1540
Coarse — compacted	24.60	15	1640

Specific gravity: fine aggregate 2.65, coarse aggregate 2.70. Readings are sample values.

Calculation

Fine aggregate (loose): 4.62/3 = 1.54 kg/l = **1540 kg/m³**

Fine aggregate (compacted): 4.92/3 = 1.64 kg/l = **1640 kg/m³**

Coarse aggregate (loose): 23.10/15 = 1.54 kg/l = **1540 kg/m³**

Voids in fine aggregate (loose) = [(2.65 × 1000 – 1540)/(2.65 × 1000)] × 100 = **41.9 %**

Voids in coarse aggregate (compact) = [(2.70 × 1000 – 1640)/(2.70 × 1000)] × 100 = **39.3 %**

Result

The bulk density of the fine aggregate is **1540 kg/m³ (loose)** and **1640 kg/m³ (compacted)**; for the coarse aggregate it is **1540 kg/m³ (loose)** and **1640 kg/m³ (compacted)**. The voids are about 42% in fine and 39% in coarse aggregate.

Precautions

- Fill the measure from a height of about 50 mm without compacting for the loose bulk density test.
- Strike off the top surface carefully so that no aggregate is lost or pressed down.
- Tamp each layer uniformly (25 strokes) and distribute the strokes over the whole surface.
- Determine the volume of the measure accurately by filling it with water.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 11

Determination of water absorption of fine and coarse aggregates (IS: 2386 Part 3)

Aim

To determine the water absorption of the given fine and coarse aggregates after 24 hours of immersion in water.

Apparatus / Equipment Required

Weighing balance (0.1 g), oven, vessel for immersion, wire basket; Absorbent cloth, tray, sample of aggregate.

Theory

Water absorption is the percentage of water absorbed by an aggregate when soaked in water for 24 hours. Porous aggregates absorb more water, which reduces the effective water available for hydration and affects the water-cement ratio.

As per IS: 2386 (Part 3), the water absorption of aggregate suitable for concrete should normally be limited to about 2% for coarse aggregate and 1–2% for fine aggregate.

Formulae Used

- Water absorption = [(W■ – W■) / W■] × 100
- where W■ = mass of oven-dry sample and W■ = mass of saturated surface-dry sample

Procedure

1. The aggregate sample was washed and dried in an oven at 100–110 °C to constant mass.
2. The oven-dry mass W■ was recorded.
3. The sample was immersed in clean water for 24 hours at room temperature.
4. The sample was taken out, the surface water removed with an absorbent cloth, and the saturated surface-dry mass W■ was recorded.
5. The water absorption was calculated and the mean of three samples taken.

Observation

Sample	Dry mass W■ (g)	SSD mass W■ (g)	Absorption (%)
Fine agg. — 1	1000	1012	1.20
Fine agg. — 2	1000	1010	1.00
Coarse agg. — 1	2000	2012	0.60
Coarse agg. — 2	2000	2016	0.80

Aggregate: 20 mm crushed stone and river sand. Readings are sample values.

Calculation

Fine aggregate: (1012 – 1000)/1000 × 100 = **1.20 %**

Coarse aggregate: (2012 – 2000)/2000 × 100 = **0.60 %**

Both values are less than the permissible limit of 2%, hence the aggregates are suitable for concrete.

Result

The water absorption of the given fine aggregate is **1.10 %** (mean) and that of the coarse aggregate is **0.70 %** (mean). Both are within the permissible limit, hence the aggregates are suitable for concrete.

Precautions

- Dry the sample to constant mass, otherwise absorption will be underestimated.
- Remove only the surface moisture with a cloth; do not dry the pores.
- Use clean water and keep the sample fully immersed during the 24 hours.
- Weigh the samples quickly to prevent evaporation.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 12

Determination of impact value of coarse aggregate (IS: 2386 Part 4)

Aim

To determine the aggregate impact value of the given coarse aggregate, which indicates its resistance to sudden shock or impact.

Apparatus / Equipment Required

Aggregate impact testing machine (with 13.5 kg hammer falling through 380 mm); IS sieves 12.5 mm and 2.36 mm, cylindrical measure, tamping rod, weighing balance, oven.

Theory

The impact value is a measure of the toughness of an aggregate and is used to judge its suitability for road and concrete work. The test consists of subjecting the aggregate to 15 blows of a standard hammer and measuring the percentage of material crushed to finer than 2.36 mm.

As per IS: 2386 (Part 4), an impact value up to 30% is satisfactory for concrete and wearing surfaces; for road surfaces the value should preferably be below 20–30%.

Formulae Used

- Impact value = (W■ / W■) × 100
- where W■ = mass of oven-dry sample and W■ = mass of material passing the 2.36 mm sieve after test

Procedure

1. The aggregate was sieved and material passing 12.5 mm and retained on 10 mm was taken and dried.
2. The measure was filled in three layers, each tamped 25 times, and the mass W■ was recorded.
3. The sample was placed in the cup of the impact machine and the hammer was released 15 times.
4. The crushed material was sieved on the 2.36 mm sieve and the passing mass W■ was weighed.
5. The impact value was calculated as a percentage of the total mass.

Observation

Trial	Mass of sample W■ (g)	Mass passing 2.36 mm W■ (g)	Impact value (%)
1	500	72	14.4
2	500	70	14.0
3	500	71	14.2
Mean	500	71.0	14.2

Aggregate: 20 mm crushed stone, 12.5 mm to 10 mm size fraction. Readings are sample values.

Calculation

Impact value = (72 / 500) × 100 = **14.4 %**

Mean impact value = (14.4 + 14.0 + 14.2)/3 = **14.2 %**

As 14.2% < 20%, the aggregate is classified as **strong / excellent** for wearing surfaces.

Result

The aggregate impact value of the given coarse aggregate is **14.2 %**, which is well within the permissible limit of 30% and indicates strong, tough aggregate suitable for concrete and road work.

Precautions

- The hammer must fall freely through exactly 380 mm and the machine must be on a level base.
- Apply exactly 15 blows at intervals of at least one second.
- Sieve the crushed sample thoroughly before weighing the passing material.
- Use a representative, oven-dry sample of the specified size fraction.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 13

Determination of crushing value of coarse aggregate (IS: 2386 Part 4)

Aim

To determine the crushing value of the given coarse aggregate, which indicates its resistance to crushing under a gradually applied compressive load.

Apparatus / Equipment Required

Aggregate crushing test apparatus (cylinder, plunger and base plate); Compression testing machine (2000 kN), IS sieves 12.5 mm and 2.36 mm, tamping rod, balance, oven.

Theory

The crushing value gives a relative measure of the resistance of an aggregate to crushing under a gradually applied compressive load. It is used for judging the suitability of aggregate for road surfaces and concrete.

As per IS: 2386 (Part 4), a crushing value of up to 30% is acceptable for concrete; for road wearing courses it should not exceed 30% and preferably 20%.

Formulae Used

- Crushing value = (W■ / W■) × 100
- where W■ = mass of the sample and W■ = mass passing the 2.36 mm sieve after crushing

Procedure

1. The aggregate passing 12.5 mm and retained on 10 mm was dried and filled in the cylinder in three layers, each tamped 25 times.
2. The mass of the sample W■ was recorded and the plunger was placed on the surface.
3. The assembly was placed in the compression testing machine and the load was applied uniformly to reach 40 tonnes in about 10 minutes.
4. The load was released and the crushed material was sieved on the 2.36 mm sieve.
5. The mass passing W■ was weighed and the crushing value calculated.

Observation

Trial	Mass of sample W■ (g)	Mass passing 2.36 mm W■ (g)	Crushing value (%)
1	3000	540	18.0
2	3000	552	18.4
3	3000	546	18.2
Mean	3000	546	18.2

Load applied: 40 tonnes in 10 minutes. Readings are sample values.

Calculation

Crushing value = (540 / 3000) × 100 = **18.0 %**

Mean crushing value = (18.0 + 18.4 + 18.2)/3 = **18.2 %**

As 18.2% < 30%, the aggregate is suitable for concrete and for road wearing courses.

Result

The crushing value of the given coarse aggregate is **18.2 %**, which is less than the permissible limit of 30%. Hence the aggregate has good crushing strength and is suitable for concrete work.

Precautions

- Apply the load uniformly and reach 40 tonnes in about 10 minutes.
- Tamp each layer 25 times uniformly over the surface.
- Use oven-dry aggregate of the specified size fraction only.
- Clean the apparatus and sieves before and after the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 14

Determination of abrasion value of coarse aggregate by Los Angeles machine (IS: 2386 Part 4)

Aim

To determine the Los Angeles abrasion value of the given coarse aggregate.

Apparatus / Equipment Required

Los Angeles abrasion testing machine (with steel balls, 500–700 revolutions); IS sieves 1.70 mm, 20 mm, 12.5 mm, 10 mm, weighing balance, oven.

Theory

The Los Angeles abrasion test measures the resistance of an aggregate to wear by abrasion, which is important for aggregates used in road surfaces and floors. The aggregate is rotated in a drum with steel balls and the loss in mass indicates its toughness and abrasion resistance.

As per IS: 2386 (Part 4), an abrasion value up to 30% is satisfactory for concrete and up to 40% for road surfaces.

Formulae Used

- Abrasion value = (loss in mass / initial mass) × 100

Procedure

1. A sample of 5000 g of the specified grading was prepared, washed and oven-dried.
2. The sample and the required number of steel balls (12 for grading A) were placed in the drum.
3. The machine was run at 30–33 revolutions per minute for 500 revolutions.
4. The material was taken out and sieved on the 1.70 mm sieve; the retained material was weighed.
5. The loss in mass was calculated and expressed as a percentage of the initial mass.

Observation

Trial	Initial mass (g)	Mass retained on 1.70 mm (g)	Loss in mass (g)	Abrasion value (%)
1	5000	3900	1100	22.0
2	5000	3925	1075	21.5
Mean	5000	3912.5	1087.5	21.8

Grading A, 12 steel balls, 500 revolutions at 30–33 rpm. Readings are sample values.

Calculation

Loss in mass = 5000 − 3900 = 1100 g

Abrasion value = (1100 / 5000) × 100 = **22.0 %**

Mean abrasion value = **21.8 %**, which is less than the 30% limit for concrete work.

Result

The Los Angeles abrasion value of the given coarse aggregate is **21.8 %**. Since it is below 30%, the aggregate has good abrasion resistance and is suitable for concrete and road work.

Precautions

- Use the correct grading and number of steel balls as per IS: 2386 (Part 4).
- Maintain the specified speed of 30–33 rpm and the exact number of revolutions.
- Wash and dry the sample before weighing to remove dust and fines.
- Ensure that the machine door is tightly closed during the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 15

Determination of workability of concrete by slump / compaction factor test (IS: 1199)

Aim

To determine the workability of the given concrete mix by the slump test (and to compare it with the compaction factor test).

Apparatus / Equipment Required

Slump cone (bottom 200 mm, top 100 mm, height 300 mm) with base plate and tamping rod (16 mm); Compaction factor apparatus, trowel, scoop, weighing balance, slump scale.

Theory

Workability describes the ease with which concrete can be mixed, transported, placed and compacted without segregation. It is commonly measured by the slump test, which indicates the consistency of the mix; the compaction factor test is used for mixes of low workability.

As per IS: 1199, a slump of 25–75 mm represents medium workability, suitable for normal RCC work; a slump of 75–125 mm is high workability and 125–200 mm is very high.

Formulae Used

- Slump = height of cone – height of the concrete after removing the cone (mm)
- Compaction factor = mass of partially compacted concrete / mass of fully compacted concrete

Procedure

1. The internal surface of the slump cone and base plate was cleaned and oiled.
2. Concrete was filled in the cone in four layers, each layer being tamped 25 times with the tamping rod.
3. The top surface was levelled and the cone was lifted vertically slowly.
4. The slump, i.e. the subsidence of the concrete, was measured with the scale.
5. For the compaction factor test, concrete was placed in the upper hopper and allowed to fall into the lower hopper and then into the cylinder, and the masses were recorded.

Observation

Test	Observation	Value
Slump — mix M25 (w/c 0.50)	Height of cone	300 mm
	Height after slump	255 mm
	Slump	45 mm
Compaction factor — partially compacted mass	—	12.30 kg
Compaction factor — fully compacted mass	—	14.60 kg
Compaction factor	12.30 / 14.60	0.84

Mix M25, cement 380 kg/m³, w/c ratio 0.50, coarse aggregate 20 mm. Readings are sample values.

Calculation

Slump = 300 – 255 = **45 mm**

Compaction factor = 12.30 / 14.60 = **0.84**

As per IS: 1199, a slump of 45 mm corresponds to medium workability, and a compaction factor of 0.84 corresponds to a mix of low–medium workability; both values agree.

Result

The slump of the given concrete mix is **45 mm** and its compaction factor is **0.84**. The mix has **medium workability**, which is suitable for reinforced concrete work.

Precautions

- Fill the cone in four equal layers and tamp each layer 25 times uniformly.
- Lift the cone vertically without any twisting or vibration.
- Measure the slump immediately after lifting the cone.
- Clean and oil the apparatus before every test to avoid friction.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 16

Determination of compressive strength of concrete cubes at 7 and 28 days (IS: 516)

Aim

To determine the compressive strength of the given concrete mix by testing 150 mm cubes at 7 days and 28 days of curing.

Apparatus / Equipment Required

150 mm cube moulds with base plates, tamping rod, vibrating table; Compression testing machine (2000 kN), curing tank, weighing balance, slump cone, trowel.

Theory

Compressive strength is the most important property of concrete and is used as a measure of its quality. Concrete cubes of 150 mm are cast, cured under water at 27 ± 2 °C and tested in a compression testing machine at the specified age.

As per IS: 516, the compressive strength is the failure load divided by the cross-sectional area of the cube. The 28-day strength is used for acceptance, while the 7-day strength (about two-thirds of the 28-day value) gives an early indication of quality.

Formulae Used

- Compressive strength, f_{ck} = P / A (N/mm²)
- where P = failure load in N and A = area of cube in mm² (150 × 150 = 22500 mm²)
- Percentage gain at 7 days ≈ (7-day strength / 28-day strength) × 100

Procedure

1. The concrete mix (M25, 1:1:2 by mass, w/c 0.50) was prepared and its workability checked by the slump test.
2. The cube moulds were cleaned, oiled and filled in three layers, each compacted by 35 strokes or on a vibrating table.
3. The top surface was levelled and the cubes were kept covered for 24 hours.
4. The cubes were demoulded and cured in water at 27 ± 2 °C until the date of testing.
5. Three cubes were tested at 7 days and three at 28 days in the compression testing machine at a uniform rate of loading of 140 kg/cm² per minute.
6. The compressive strength of each cube and the mean strength were calculated.

Observation

Age	Cube No.	Failure load (kN)	Area (mm²)	Strength (N/mm²)
7 days	1	430	22500	19.11
7 days	2	445	22500	19.78
7 days	3	438	22500	19.47
28 days	1	665	22500	29.56
28 days	2	672	22500	29.87
28 days	3	668	22500	29.69

Mix M25; w/c ratio 0.50; slump 45 mm. Area of 150 mm cube = 22500 mm². Readings are sample values.

Calculation

7-day mean strength = (19.11 + 19.78 + 19.47)/3 = **19.45 N/mm²**

28-day mean strength = (29.56 + 29.87 + 29.69)/3 = **29.71 N/mm²**

Sample calculation: 430 kN / 22500 mm² = 430000/22500 = 19.11 N/mm²

Percentage gain at 7 days = (19.45/29.71) × 100 = **65.5 %**, which is normal.

For M25 concrete, the target mean strength at 28 days is 25 + 1.65 × 4 = 31.6 N/mm²; the observed value of 29.71 N/mm² is about 94% of the target and requires slightly better control of the mix.

Result

The compressive strength of the given concrete is **19.45 N/mm² at 7 days** and **29.71 N/mm² at 28 days**. The 7-day strength is about 65% of the 28-day strength, showing normal strength development of the mix.

Precautions

- Compact the concrete uniformly; under-compaction reduces strength considerably.
- Keep the cube moulds on a level, rigid surface and do not disturb them during initial setting.
- Cure the cubes in water at 27 ± 2 °C until testing; do not allow them to dry.
- Test the cube with the loading faces parallel and at right angles to the compacted direction.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 17

Assessment of durability of concrete by Rapid Chloride Penetration Test / permeability test

Aim

To assess the durability of the given concrete by measuring its resistance to chloride ion penetration using the Rapid Chloride Penetration Test (RCPT) as per ASTM C 1202.

Apparatus / Equipment Required

RCPT apparatus with two cells, DC power supply (60 V), data logger; Concrete disc specimen (100 mm diameter, 50 mm thick), vacuum saturation unit, 3% NaCl solution, 0.3 N NaOH solution.

Theory

Durability of concrete depends mainly on its permeability, i.e. the ease with which water and aggressive ions can enter the concrete. The RCPT measures the electrical charge passed through a saturated concrete specimen in 6 hours under 60 V DC; a lower charge indicates a more impermeable and durable concrete.

As per ASTM C 1202, the charge passed is classified as: >4000 coulombs — high permeability; 2000–4000 — moderate; 1000–2000 — low; 100–1000 — very low; <100 — negligible.

Formulae Used

- Charge passed, $Q = \int I \, dt$ (coulombs, integrated over 6 hours)
- Total charge Q = current (amperes) × time (seconds)

Procedure

1. A 100 mm diameter core or disc specimen of 50 mm thickness was cut from the concrete sample.
2. The specimen was vacuum-saturated with water and then placed between the two cells of the apparatus.
3. The cell on one face was filled with 3% NaCl solution (cathode) and the other with 0.3 N NaOH solution (anode).
4. A potential difference of 60 V DC was applied for 6 hours and the current was recorded every 30 minutes.
5. The total charge passed was computed by integrating the current-time curve.

Observation

Time (min)	Current (mA)	Charge increment (C)	Cumulative charge (C)
30	78	140	140
60	82	144	284
120	85	306	689
240	88	622	1580
360	90	640	1800

Specimen: M25 concrete, 28 days cured. Charge increment = average current × time interval.

Calculation

Total charge passed in 6 hours = **1800 coulombs**

Sample increment: average current 85 mA for 60 minutes → Q = 0.085 × 3600 = 306 C

As per ASTM C 1202, 1800 C lies in the range 1000–2000 C, which indicates **low permeability**.

Result

The charge passed through the concrete specimen is **1800 coulombs**, which corresponds to **low chloride ion permeability** as per ASTM C 1202. Hence the concrete has good durability against chloride attack.

Precautions

- The specimen must be fully saturated; dry specimens give misleading low values.
- Maintain the applied voltage at exactly 60 V throughout the test.
- Seal the specimen properly in the cells to prevent leakage between the solutions.
- Handle the specimen carefully — cracks and broken edges increase the charge passed.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 18

Determination of strength of hardened concrete by non-destructive tests (rebound hammer and ultrasonic pulse velocity)

Aim

To assess the in-situ strength and quality of hardened concrete using the rebound hammer test and the ultrasonic pulse velocity test.

Apparatus / Equipment Required

Schmidt rebound hammer (concrete test hammer) with calibration anvil; Ultrasonic pulse velocity apparatus with transducers, coupling gel, measuring tape.

Theory

Non-destructive tests (NDT) are used to assess the quality of concrete in existing structures without damaging them. The rebound hammer measures surface hardness, which is correlated with compressive strength through a calibration curve. The ultrasonic pulse velocity test measures the velocity of an ultrasonic pulse through concrete, which depends on its density and uniformity.

As per IS: 13311, pulse velocity above 4.5 km/s indicates excellent quality, 3.5–4.5 km/s good, 3.0–3.5 km/s medium, and below 3.0 km/s doubtful concrete.

Formulae Used

- Rebound number = average of the rebound readings
- Pulse velocity, V = L / t (km/s), where L = path length in mm and t = travel time in µs
- Compressive strength is read from the calibration curve supplied with the rebound hammer.

Procedure

1. The concrete surface was cleaned, rubbed smooth and divided into a grid of test points.
2. The rebound hammer was pressed firmly against the surface and the plunger was released; ten readings were taken at different points and the mean rebound number was calculated.
3. For the pulse velocity test, coupling gel was applied to the transducers and they were placed on opposite faces of the member.
4. The travel time was measured and the path length was measured with a tape.
5. The pulse velocity was calculated and the compressive strength was estimated from the calibration charts.

Observation

Test	Observation	Mean value
Rebound number (10 readings)	34, 36, 35, 33, 37, 35, 34, 36, 35, 35	35.0
Path length, L	—	300 mm
Travel time, t	—	72 µs
Pulse velocity, V	300/72	4.17 km/s

Member tested: RCC column, 28 days old. Readings are sample values.

Calculation

Mean rebound number = (34+36+35+33+37+35+34+36+35+35)/10 = **35.0**

From the rebound hammer calibration curve for a horizontal surface, a rebound number of 35 corresponds to about **27–28 N/mm²**.

Pulse velocity = 300 / 72 = **4.17 km/s**

As per IS: 13311, a velocity of 4.17 km/s (3.5–4.5 km/s) indicates **good quality** concrete.

Result

The average rebound number of the concrete is **35**, indicating a compressive strength of about **27.5 N/mm²**, and the ultrasonic pulse velocity is **4.17 km/s**. Both results indicate **good quality, uniform concrete** with no visible internal defects.

Precautions

- Test only a smooth, dry and clean surface; rough or plastered surfaces give low readings.
- Hold the hammer perpendicular to the surface and take at least 10 readings at different points.
- Apply sufficient coupling gel for the pulse velocity test and ensure good contact between transducer and concrete.
- Measure the path length accurately and avoid testing near edges, cracks or reinforcement.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	M.S. Shetty, *Concrete Technology*, S. Chand & Co., New Delhi.
2	M.L. Gambhir, *Concrete Technology*, McGraw Hill Education.
3	A.M. Neville, *Properties of Concrete*, Pearson Education.
4	IS: 4031 (Parts 1–6) — Methods of physical tests for hydraulic cement.
5	IS: 2386 (Parts 1–4) — Methods of test for aggregates for concrete.
6	IS: 1199, IS: 516, IS: 10262, IS: 456 — workability, strength, mix design and concrete code.
7	IS: 13311 (Parts 1 & 2), IS: 3085 and ASTM C 1202 — non-destructive and permeability tests.
8	SBTE Bihar curriculum for Concrete Technology (2415302 / P2415302).

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.

STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

BASIC SURVEYING (CE, AE)

Course Code: 2415301 (P2415301)

Diploma in Civil Engineering • Semester III

Student's Name	
Enrollment No.	
Roll No.	
Branch	Civil Engineering
Semester	III
Session	20____ – 20____
Name of Institute	

Department of Civil Engineering

(Name & address of the institute)

How to write this file

Is file ke har practical ko nimn likhit format mein likha gaya hai. Apne college ke lab manual se practical ki sankhya aur readings confirm karke observation table mein apni actual readings bharein — yahan di gayi readings sample hain, sirf samajhne ke liye.

- **Aim** — practical ka uddeshya ek line mein.
- **Apparatus** — naam aur specification (capacity, IS code).
- **Theory / Formula** — principle aur formula jo calculation mein use hua.
- **Procedure** — step-wise, past tense mein, chhote vakya.
- **Observation table** — readings, units ke saath; IS code ka reference.
- **Calculation** — formula lagakar steps dikhayein.
- **Result** — answer with unit aur IS permissible limit ke saath.
- **Precautions** — 3 se 4 points.

General instructions

- Har practical ke baad date aur teacher ke signature zaroor lein.
- Diagram pencil se, scale se banayein; likhai blue/black pen se.
- Index page mein page number bharna na bhoolein — file check ke waqt yahi dekha jata hai.
- Practical exam (ELA) mein viva hota hai — aim, formula aur result yaad rakhein.

Index

Practical exam (ELA) mein viva hota hai — har practical ka aim, formula aur result yaad rakhein. Page number index mein khud bhar lein (ya teacher ke kehne par).

S. No.	Title of the Practical	Page No.	Date	Signature
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09	Differential levelling — determination of reduced levels by Height of Instrument and Rise and Fall methods	12		
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PRACTICAL NO. 01

Measurement of distance between two inter-visible stations using chain, tape and ranging rods

Aim

To measure the horizontal distance between two inter-visible survey stations A and B with a 20 m chain, a metallic tape and ranging rods, and to compare the two measurements.

Apparatus / Equipment Required

20 m metric chain (100 links) with brass handles; 30 m metallic tape; Ranging rods (2 m) — 3 nos; Arrows (pegs) — 10 nos., plumb bob, cross staff, field book and pencil.

Theory

Chaining is the process of measuring the horizontal distance between two points with a chain or tape. Before chaining, the line AB is made straight by ranging, i.e. by fixing intermediate points exactly in line with the end stations using ranging rods.

Direct ranging is used when the two stations are inter-visible and the ground is fairly level. The follower holds the zero end of the chain at A and the leader drags the chain towards B, inserting an arrow at every chain length. The last portion is measured with the tape or by counting links.

Procedure

1. Station A and station B were marked with pegs and ranging rods were fixed vertically at both ends.
2. Both ranging rods were sighted and intermediate points were fixed at every 20 m so that all points lie in one straight line.
3. The zero end of the chain was held at A and the chain was stretched along the ranged line; an arrow was fixed at the end of each chain length.
4. The number of full chain lengths and the remaining links/tape length were recorded in the field book.
5. The same distance was measured independently with the metallic tape.
6. The measurement was repeated three times in each case and the mean value was calculated.

Observation

Trial No.	Chain measurement (m)	Tape measurement (m)	Difference (m)
1	24.380	24.372	0.008
2	24.365	24.360	0.005
3	24.375	24.368	0.007
Mean	24.373	24.367	0.006

Average chain length = 20.0 m; average tape temperature ≈ 30 °C. Readings are sample values.

Calculation

Mean chain distance = (24.380 + 24.365 + 24.375) / 3 = **24.373 m**

Mean tape distance = (24.372 + 24.360 + 24.368) / 3 = **24.367 m**

Difference = 24.373 − 24.367 = 0.006 m = 6 mm, which is within permissible chaining error (1 in 1000 for ordinary chaining, i.e. about 24 mm for 24 m).

Result

The distance between stations A and B is **24.37 m** (chain) and **24.37 m** (tape); the difference of 6 mm is within permissible limits and the measurement is accepted.

Precautions

- The chain must be perfectly straight and free from kinks before use.
- Ranging rods should be held truly vertical; check with a plumb bob on sloping ground.
- Arrows must be inserted exactly at the chain end and counted at the end of the day.
- Care should be taken to apply a uniform pull; dragging the chain over hard surfaces damages it.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 02

Reciprocal ranging and measurement of distance between two non-inter-visible stations

Aim

To carry out reciprocal ranging between two stations A and B which are not inter-visible due to a rising ground/hill, and to measure the horizontal distance between them.

Apparatus / Equipment Required

Ranging rods — 4 nos; 20 m chain, 30 m tape, arrows, plumb bob, field book.

Theory

When two end stations are not inter-visible because of an intervening obstruction such as a hillock or a building, direct ranging is not possible. In such a case two intermediate points are selected and shifted alternately until both lie on the line AB — this is called reciprocal or indirect ranging.

Two chainmen stand at M1 and N1, each sighting the other and the far station. The point nearer to the line is shifted successively until the four points A, M, N, B are exactly in one line. Thereafter the distance is measured in two parts AM and MB (or AN and NB).

Procedure

1. Ranging rods were fixed at stations A and B and the obstruction was located.
2. Two points M1 and N1 were selected on either side of the obstruction, roughly in line with AB.
3. The person at M1 directed the person at N1 to move sideways until N1 appeared in line with M1 and B.
4. The person at N1 then directed the person at M1 to shift until M1 appeared in line with N1 and A.
5. The process was repeated until both M and N were exactly on the line AB.
6. The distance was measured as AM and MB separately with the chain and the total AB was obtained.

Observation

Portion measured	No. of full chains	Remaining (m)	Distance (m)
A to M	0	15.420	15.420
M to B	0	18.235	18.235
Total AB	—	—	33.655

Readings are sample values; the intermediate point was near the top of the obstruction.

Calculation

AB = AM + MB = 15.420 + 18.235 = **33.655 m**

Check: repeat measurement in reverse direction = 33.650 m; difference = 0.005 m = 5 mm, well within the permissible limit of 1 in 1000.

Result

The distance between the non-inter-visible stations A and B is **33.655 m** and the points A, M, N, B were found to be exactly in one straight line.

Precautions

- Ranging rods must be held vertical and exactly on the peg point.
- Both chainmen should sight the rods with one eye closed to avoid parallax error.
- The shifting should be done in small steps, otherwise the process does not converge.
- The chain must not be lifted while dragging over uneven ground.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 03

Determination of area of an open field by chain and cross-staff survey

Aim

To determine the area of a given open field by measuring a base line and taking offsets with a cross-staff at regular intervals.

Apparatus / Equipment Required

Cross-staff (with optical square), ranging rods, 20 m chain, 30 m tape; Arrows, plumb bob, field book, drawing sheet and scale.

Theory

In chain survey, the area of an irregular field is found by dividing it into triangles and trapezoids. A base line is run through the field and perpendicular offsets are measured to the boundary at regular intervals. The area is then computed by the trapezoidal rule or by dividing the figure into triangles.

Trapezoidal rule: $A = d\,[\,(O_1 + O_n)/2 + O_2 + O_3 + \ldots + O_{n-1}\,]$, where d is the equal interval between offsets and O are the offset lengths.

Formulae Used

- Area (trapezoidal rule) = $d \times [\,(O_1 + O_n)/2 + \Sigma\,(\text{intermediate offsets})\,]$
- Area of a triangle = ½ × base × height

Procedure

1. A base line AB was selected through the middle of the field and station pegs were fixed at both ends.
2. The line was ranged and chained; the total length was found to be 40 m.
3. Offsets were taken on both sides at intervals of 10 m using a cross-staff, the perpendicularity being checked by the optical square.
4. The chainage and offset readings were entered in the field book in the tabular form.
5. The field book was plotted to a suitable scale on a drawing sheet.
6. The area was calculated by the trapezoidal rule and verified by counting squares/graphically.

Observation

Chainage (m)	Offset to left (m)	Offset to right (m)	Mean offset (m)
0 (A)	0.00	0.00	0.00
10	8.00	6.20	7.10
20	10.00	8.40	9.20
30	6.40	5.60	6.00
40 (B)	0.00	0.00	0.00

Interval d = 10 m. Offsets were measured on both sides of the base line.

Calculation

Area on left = $10 \times [\,(0 + 0)/2 + 8.00 + 10.00 + 6.40\,] = 10 \times 24.40 =$ **244.0 m²**

Area on right = $10 \times [\,(0 + 0)/2 + 6.20 + 8.40 + 5.60\,] = 10 \times 20.20 =$ **202.0 m²**

Total area = 244.0 + 202.0 = **446.0 m²** = 0.0446 hectare

Check by plotting: area measured with planimeter on the plotted sheet = 4.45 cm² at 1:100 scale = 445 m², which agrees within 1%.

Result

The area of the given open field is **446 m²** (0.0446 hectare).

Precautions

- Offsets must be taken exactly perpendicular to the base line; check with optical square.
- The base line should be chosen on the longest available line through the field.
- Cross-staff must be held vertical and readings taken at eye level.
- All measurements should be entered in the field book at the site itself, never from memory.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 04

Measurement of fore bearing and back bearing of survey lines of an open traverse using prismatic compass

Aim

To measure the fore bearing and back bearing of the lines of an open traverse with a prismatic compass and to check the difference between them.

Apparatus / Equipment Required

Prismatic compass with tripod stand; Ranging rods, tape, pegs, field book, pencil, plumb bob.

Theory

The prismatic compass is a magnetic compass in which the graduated ring is attached to the needle and rotates with it; the reading is taken through a prism while sighting the object. Bearings are measured clockwise from the magnetic north.

Fore bearing (FB) is the bearing of a line taken in the direction of the survey; back bearing (BB) is the bearing taken in the opposite direction. For a line free from local attraction, BB = FB ± 180°. Any difference from 180° indicates either local attraction or an observational error.

Formulae Used

- BB = FB + 180° (if FB < 180°) or BB = FB – 180° (if FB > 180°)

Procedure

1. The compass was set up over station A, levelled with the ball and socket arrangement, and the magnetic needle was allowed to settle.
2. The line AB was sighted through the prism and the fore bearing was read and recorded.
3. The compass was then taken to station B and the back bearing of AB was measured by sighting A.
4. In the same manner the compass was shifted to successive stations and FB/BB of each line were recorded.
5. The difference between FB and BB of every line was worked out and compared with 180°.

Observation

Line	Fore bearing (FB)	Back bearing (BB)	Difference (FB – BB)	Remark
AB	42°30′	222°30′	180°00′	Correct
BC	96°00′	275°30′	179°30′	Error 30′
CD	152°15′	332°15′	180°00′	Correct
DE	236°45′	56°45′	180°00′	Correct

Readings are sample values. Line BC shows a discrepancy of 30′, which indicates local attraction at station B or C.

Calculation

For line AB: FB = 42°30′, BB should be 42°30′ + 180° = 222°30′ — observed 222°30′, hence no local attraction.

For line BC: FB = 96°00′, expected BB = 276°00′, observed 275°30′ — discrepancy of 30′.

For line CD: FB = 152°15′, expected BB = 332°15′ — correct.

For line DE: FB = 236°45′, expected BB = 56°45′ — correct.

Since three lines are correct, the error of 30′ is due to local attraction at station B (or C); all bearings taken at that station should be corrected by ±30′.

Result

The fore and back bearings of all the lines of the open traverse were measured. Lines AB, CD and DE are free from local attraction; line BC shows a difference of 30′, which is attributed to local attraction near station B.

Precautions

- Remove metal objects such as watch, keys, chain and steel tape from near the compass.
- Do not take readings under overhead electric lines or near magnetic material.
- Always read the bearing with the object exactly bisected by the vertical hair.
- Note the local time and date; the magnetic declination must be applied if true bearings are required.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 05

Closed traverse of five sides by prismatic compass — correction of bearings for local attraction and calculation of included angles

Aim

To measure the fore and back bearings of a closed traverse of five sides, to correct the bearings for local attraction, and to compute the included angles and check their sum.

Apparatus / Equipment Required

Prismatic compass with tripod; Ranging rods, tape, pegs (5 nos.), field book, protractor and drawing sheet.

Theory

In a closed traverse the difference between the fore and back bearing of every line must be exactly 180°. If for any line it is not 180°, one or both of its stations are affected by local attraction. Lines whose FB and BB differ by exactly 180° are free from local attraction, and corrections for the remaining stations are worked out from these lines. The included angle at a station is the back bearing of the preceding line minus the fore bearing of the next line, and the sum of the interior angles of an n-sided closed traverse is (2n – 4) × 90°, i.e. 540° for a pentagon.

Formulae Used

- BB = FB ± 180°
- Interior angle at station = BB (previous line) – FB (next line) [+ 360° if negative]
- Σ interior angles = (2n – 4) × 90° = 540° for n = 5

Procedure

1. Five stations A, B, C, D and E were fixed forming a closed traverse around the given area and pegs were driven.
2. The compass was set at station A and the bearing of AB was measured as FB, then at B the BB of AB was measured.
3. The instrument was shifted successively to C, D and E and the FB and BB of every line were observed.
4. The differences between FB and BB were computed to identify the station affected by local attraction.
5. Corrections were applied to the bearings observed at the affected station.
6. The included angles were calculated from the corrected bearings and their sum was checked against 540°.

Observation

Line	Observed FB	Observed BB	FB – BB	Correct FB	Correct BB
AB	60°00′	240°00′	180°00′	60°00′	240°00′
BC	132°00′	312°00′	180°00′	132°00′	312°00′
CD	204°00′	26°00′	178°00′	204°00′	24°00′
DE	278°00′	96°00′	182°00′	276°00′	96°00′
EA	348°00′	168°00′	180°00′	348°00′	168°00′

Lines AB, BC and EA show a difference of exactly 180°, hence stations A, B, C and E are free from local attraction. Station D is affected by +2°, so all bearings observed at D are reduced by 2°.

Calculation

Correction at D = –2°; corrected BB of CD = 26°00′ – 2° = 24°00′ and corrected FB of DE = 278°00′ – 2° = 276°00′.

Angle A = BB of EA – FB of AB = 168° – 60° = **108°**

Angle B = BB of AB – FB of BC = 240° – 132° = **108°**

Angle C = BB of BC – FB of CD = 312° – 204° = **108°**

Angle D = BB of CD – FB of DE = 24° + 360° – 276° = **108°**

Angle E = BB of DE – FB of EA = 96° + 360° – 348° = **108°**

Sum = 108° × 5 = **540°** = (2 × 5 – 4) × 90° — hence the traverse is closed and the corrections are correct.

Result

The local attraction was found only at station D (+2°) and was corrected. The included angles are A = B = C = D = E = 108°, and their sum is 540°, which is equal to the theoretical value for a five-sided closed traverse.

Precautions

- Keep the compass away from iron and steel objects, mobile phones and magnetic material.
- The same instrument and observer should be used throughout the traverse.
- Read bearings to the nearest 30′ or better, and take two readings at each station.
- Check the closure of the traverse before leaving the field so that any gross error can be re-measured.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 06

Survey project — chain and compass closed traverse around a building (minimum five sides)

Aim

To carry out a chain and compass survey of a closed traverse of five sides around a building and to prepare the field book for plotting.

Apparatus / Equipment Required

Prismatic compass, 20 m chain, 30 m tape, ranging rods; Pegs, arrows, plumb bob, field book, chalk for marking.

Theory

A survey project combines linear (chain) and angular (compass) measurements to fix the position of points around an obstruction such as a building. The traverse is run around the building, keeping all lines clear of the structure, and each line is measured in length as well as bearing.

For plotting, the length and bearing of each line are converted into latitude and departure: Latitude = L cos θ and Departure = L sin θ, where θ is the reduced bearing. The algebraic sums of latitudes and departures should be zero for a closed traverse; the residual is the closing error, which is distributed by Bowditch's rule.

Formulae Used

- Latitude = L cos θ ; Departure = L sin θ
- Closing error = $\sqrt{(\Sigma L)^2 + (\Sigma D)^2}$
- Correction in latitude/departure = total error × (length of line / perimeter)

Procedure

1. The building was inspected and five traverse stations A, B, C, D and E were fixed around it, clear of the walls.
2. The compass was set up at A, levelled and centred, and the bearing of AB was observed. The line AB was then chained.
3. The instrument was shifted to B and the FB of BC together with the BB of AB was recorded; the line BC was measured.
4. The process was repeated for lines CD, DE and EA and all readings were entered in the field book.
5. The FB–BB differences were checked in the field and any line showing a large discrepancy was re-measured.
6. The traverse was plotted on a drawing sheet to a scale of 1:200 and the closing error was adjusted graphically.

Observation

Line	Length (m)	FB	BB	Difference
AB	24.30	30°00′	210°00′	180°00′
BC	18.55	118°00′	298°00′	180°00′
CD	27.10	196°30′	16°30′	180°00′
DE	16.40	264°00′	84°00′	180°00′
EA	21.25	330°30′	150°30′	180°00′

Perimeter = 24.30 + 18.55 + 27.10 + 16.40 + 21.25 = 107.60 m. Readings are sample values.

Calculation

Perimeter of the traverse = **107.60 m**

Plotting scale selected: 1:200, so that the longest side (27.10 m) occupies 135.5 mm on an A1 sheet.

Sum of observed interior angles = 540°00′ (check with the theoretical value), hence no angular error.

Closing error obtained after plotting = 0.06 m in 107.60 m, i.e. about 1 in 1790, which is within the permissible limit of 1 in 1000 for a compass traverse.

The closing error was distributed by Bowditch's rule and the traverse was redrawn.

Result

The chain and compass traverse of five sides around the building was completed with a perimeter of **107.60 m**; the closing error was 0.06 m (1 in 1790), which is within permissible limits, and the corrected traverse was plotted to a scale of 1:200.

Precautions

- Keep the traverse lines clear of the building and of iron grills or gates that cause local attraction.
- Measure each line twice — forward and backward — and record the mean.
- Fix the stations permanently so that they can be re-occupied if required.
- Plot the traverse the same day while the field book is fresh.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 07

Plotting of the chain and compass traverse on an A1 size imperial drawing sheet

Aim

To plot the traverse data collected in Practical No. 6 on an A1 size imperial drawing sheet to a suitable scale and to adjust the closing error.

Apparatus / Equipment Required

A1 size imperial drawing sheet, drawing board, T-square, set squares; Protractor, scale (engineer's), drawing pins, sharp pencil, eraser, tracing paper.

Theory

Plotting is the graphical representation of the field data. The scale is chosen so that the whole traverse fits on the sheet with a margin: RF = length of drawing / length on ground. For a site of about 100 m on an A1 sheet (594 × 841 mm), a scale of 1:200 is convenient.

Each line is drawn with the help of its bearing (measured with a protractor from the magnetic meridian) and its length (scaled from the engineer's scale). A closing error, if any, is adjusted by shifting the plotted stations proportionally — graphically, this is done by drawing a closing line and dividing the error in proportion to the perimeter.

Formulae Used

- Representative fraction, RF = distance on paper / distance on ground
- Scale for the sheet: Scale = drawing length / ground length

Procedure

1. The drawing sheet was fixed on the board and a border with a title block was drawn.
2. A north line was drawn at station A and the magnetic meridian was marked with the help of the protractor.
3. The bearing of AB was set out from the meridian and the length was scaled; point B was marked.
4. The same procedure was repeated for BC, CD, DE and EA using a parallel ruler/protractor at each station.
5. The end of the last line (A′) was compared with the starting point A and the closing error AA′ was measured.
6. The closing error was distributed in proportion to the length of the lines and the corrected traverse was inked.
7. The building outline, north arrow, scale, title and legend were added to complete the sheet.

Observation

Line	Ground length (m)	Scale 1:200 (mm)	Bearing	Remarks
AB	24.30	121.5	30°00′	Plotted
BC	18.55	92.8	118°00′	Plotted
CD	27.10	135.5	196°30′	Plotted
DE	16.40	82.0	264°00′	Plotted
EA	21.25	106.3	330°30′	Plotted

A1 imperial sheet = 594 mm × 841 mm. Title block shows name, roll number, scale and date.

Calculation

RF = 1:200, hence 1 m on ground = 5 mm on paper.

AB = 24.30 m × 5 = 121.5 mm; BC = 18.55 × 5 = 92.8 mm; CD = 27.10 × 5 = 135.5 mm;

DE = 16.40 × 5 = 82.0 mm; EA = 21.25 × 5 = 106.3 mm.

Closing error AA′ measured on the sheet = 0.3 mm, i.e. 0.06 m on ground — within permissible limit.

Result

The traverse was plotted to a scale of 1:200 on an A1 size imperial sheet; the closing error was 0.06 m on ground (0.3 mm on the sheet) and was adjusted. The completed sheet shows the building outline, north line, scale and title block.

Precautions

- Use a sharp HB pencil for plotting and a 2H pencil for construction lines.
- Set every bearing from the north line at each station, not from the previous line.
- Keep the protractor centred exactly on the station point to avoid angular error.
- Do not fold the A1 sheet; roll it and submit it in a tube.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 08

Simple levelling with dumpy level / auto level to determine reduced levels of given points

Aim

To determine the reduced levels (RL) of a few given points with respect to a bench mark using a dumpy level (or auto level) and a levelling staff by simple levelling.

Apparatus / Equipment Required

Dumpy level / auto level with tripod; Levelling staff (self-reading, 4 m); Pegs, arrows, field book, umbrella.

Theory

Levelling is the operation of finding the relative elevations of points on or below the surface of the earth. The level establishes a horizontal line of collimation, and the staff readings taken on it give the vertical distances of the points from that line.

Simple levelling is used when the instrument can be set up between two points without any change of station: RL of the new point = HI – staff reading, where HI (height of instrument) = RL of bench mark + back sight.

Formulae Used

- Height of instrument, HI = RL of BM + Back sight (BS)
- RL of any point = HI – staff reading at that point

Procedure

1. The level was set up roughly midway between the bench mark and the points to be levelled and levelled with the foot screws.
2. The temporary adjustments — setting, levelling and focusing — were carried out.
3. A back sight was taken on the bench mark of known RL and recorded.
4. Intermediate sights were taken on points P, Q and R by turning the telescope only in the horizontal plane.
5. The HI was computed and the RL of every point was calculated.
6. The arithmetic check (ΣBS – ΣFS = last RL – first RL) was applied.

Observation

Station	Staff reading (m)	Type of sight	HI (m)	RL (m)	Remarks
BM	1.455	BS	101.455	100.000	Bench mark
P	1.720	IS	101.455	99.735	
Q	2.105	IS	101.455	99.350	Lowest point
R	1.980	IS	101.455	99.475	

RL of bench mark = 100.000 m (assumed). Readings are sample values.

Calculation

HI = RL of BM + BS = 100.000 + 1.455 = **101.455 m**

RL of P = 101.455 – 1.720 = **99.735 m**

RL of Q = 101.455 – 2.105 = **99.350 m**

RL of R = 101.455 – 1.980 = **99.475 m**

Check: ΣBS – ΣFS = 1.455 – 0 = 1.455; last RL – first RL cannot be applied for IS readings, hence the HI is verified by comparing the difference of RLs with the difference of staff readings (2.105 – 1.455 = 0.650 = 100.000 – 99.350 ✓).

Result

The reduced levels are: P = **99.735 m**, Q = **99.350 m**, R = **99.475 m** with respect to the bench mark of RL 100.000 m. Point Q is the lowest point of the area.

Precautions

- The level must be set up on firm ground and the tripod legs pressed down well.
- Always take the back sight first and the fore sight last at every setting.
- Keep the staff truly vertical; use a bubble in the staff or wave it slightly to get the minimum reading.
- Protect the instrument from direct sun with an umbrella to avoid expansion errors.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 09

Differential levelling — determination of reduced levels by Height of Instrument and Rise and Fall methods

Aim

To determine the reduced levels of points along a profile by differential levelling and to reduce the levels by (i) the Height of Instrument method and (ii) the Rise and Fall method.

Apparatus / Equipment Required

Dumpy level / auto level with tripod, levelling staff; Pegs, arrows, field book, pencil, umbrella.

Theory

Differential levelling is used when the points are far apart or the difference of level is large, so that the instrument has to be shifted several times; the point on which both a fore sight and a back sight are taken is called a change point. In the HI method the RL is obtained by subtracting the staff reading from the height of instrument, while in the Rise and Fall method the rise or fall between successive readings is added to or subtracted from the previous RL.

Formulae Used

- HI = RL + BS ; RL = HI – IS or FS
- Rise = previous reading – next reading (positive) ; Fall = next reading – previous reading (positive)
- Arithmetic check: ΣBS – ΣFS = ΣRise – ΣFall = last RL – first RL

Procedure

1. The level was set up at a convenient position and the first back sight was taken on the bench mark.
2. Intermediate and fore sights were taken on the points in the line; the fore sight point was used as a change point.
3. The instrument was shifted beyond the change point and a new back sight was taken on the same point.
4. The process was repeated until the last point was reached.
5. The RLs were computed first by the HI method and then independently by the Rise and Fall method.
6. The arithmetic checks were applied to both methods and the results were compared.

Observation

Station	BS (m)	IS (m)	FS (m)	HI (m)	RL (m)	Remarks
BM	1.560	—	—	101.560	100.000	Bench mark
A	—	1.230	—	101.560	100.330	
CP1	1.860	—	2.150	101.270	99.410	Change point
B	—	1.410	—	101.270	99.860	
CP2	2.010	—	2.340	100.940	98.930	Change point
C	—	—	1.480	100.940	99.460	Last point

ΣBS = 5.430 m, ΣFS = 5.970 m. Readings are sample values.

Calculation

HI_1 = 100.000 + 1.560 = 101.560 m; RL of A = 101.560 – 1.230 = **100.330 m**; RL of CP1 = 101.560 – 2.150 = **99.410 m**

HI_2 = 99.410 + 1.860 = 101.270 m; RL of B = 101.270 – 1.410 = **99.860 m**; RL of CP2 = 101.270 – 2.340 = **98.930 m**

HI_3 = 98.930 + 2.010 = 100.940 m; RL of C = 100.940 – 1.480 = **99.460 m**

Check: ΣBS – ΣFS = 5.430 – 5.970 = –0.540 m; last RL – first RL = 99.460 – 100.000 = –0.540 m ✓

By Rise and Fall method: BM → A rise 0.330 (100.330); A → CP1 fall 0.920 (99.410); CP1 → B rise 0.450 (99.860); B → CP2 fall 0.930 (98.930); CP2 → C rise 0.530 (99.460).

Check: ΣRise – ΣFall = (0.330 + 0.450 + 0.530) – (0.920 + 0.930) = 1.310 – 1.850 = –0.540 m ✓ — the same as ΣBS – ΣFS.

Result

The reduced levels obtained are A = 100.330 m, CP1 = 99.410 m, B = 99.860 m, CP2 = 98.930 m and C = 99.460 m. Both reduction methods give identical results and the arithmetic checks are satisfied.

Precautions

- A change point must be a firm, permanent object; the staff must not be lifted from it between the two settings.
- Take the fore sight and the next back sight on the same point before shifting the instrument.
- Avoid very long sights and keep the backsight and foresight distances nearly equal.
- Enter readings in the field book immediately and never rub out a wrong entry — strike it out and rewrite.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 10

Fly levelling with double check using dumpy level / auto level

Aim

To carry out fly levelling from a bench mark to a distant point and to check the accuracy by levelling back to the starting bench mark (double check).

Apparatus / Equipment Required

Dumpy level / auto level with tripod, levelling staff; Pegs, arrows, field book, umbrella, measuring tape.

Theory

Fly levelling is a quick method of differential levelling used to connect a distant point to a bench mark; intermediate sights are avoided and only back sights and fore sights are taken. Since the instrument is shifted many times, the total error accumulates, and it is therefore checked by running the levelling back to the same bench mark.

The difference between the forward and backward differences of level is the closing error (misclosure). The permissible closing error for ordinary levelling is about $\pm 5\sqrt{K}$ mm, where K is the distance in km.

Formulae Used

- Difference of level = $\Sigma BS - \Sigma FS$
- Closing error = (forward difference of level) – (backward difference of level)
- Permissible error = $\pm\, 5\ \sqrt{K}$ mm (K in km)

Procedure

1. The level was set up near the bench mark and a back sight was taken.
2. The instrument was shifted forward and a fore sight was taken on a firm point before shifting, which became the new back sight.
3. Only back sights and fore sights were taken until the destination point was reached, and the forward difference of level was computed.
4. The levelling was then repeated in the reverse direction from the destination back to the bench mark.
5. The backward difference of level was computed and compared with the forward value.
6. The closing error was worked out and compared with the permissible limit.

Observation

Leg of run	ΣBS (m)	ΣFS (m)	Difference of level (m)	Remarks
Forward run (BM → P)	8.465	9.710	–1.245	Fall of 1.245 m
Backward run (P → BM)	9.706	8.461	+1.245	Rise of 1.245 m
Mean	—	—	–1.245	Accepted value

Total length of the run = 0.60 km. Readings are sample values.

Calculation

Forward run: 8.465 – 9.710 = –**1.245 m** (point P is 1.245 m below the bench mark).

Backward run: 9.706 – 8.461 = **+1.245 m** (BM is 1.245 m above P).

Misclosure = 1.245 – 1.245 = **0.000 m** (in the sample data); for field data a small misclosure of 3–5 mm is normal.

Permissible error = $\pm 5\sqrt{0.60} = \pm 3.9$ mm ≈ **±4 mm**. A misclosure within this value is acceptable.

Result

Fly levelling was completed over a run of 0.60 km. The difference of level between the bench mark and point P is **1.245 m** (P below BM) and the closing error is within the permissible limit of ±4 mm, hence the work is accepted.

Precautions

- Keep the lengths of back sight and fore sight nearly equal at each setting to cancel collimation error.
- Never take an intermediate sight in fly levelling; only BS and FS are recorded.
- Use a firm and stable change point and take the reading without lifting the staff.
- Complete both forward and backward runs on the same day under similar atmospheric conditions.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 11

Survey project — contour map of a 150 m × 150 m block by block contouring (grid 10 m × 10 m)

Aim

To prepare a contour map of a given 150 m × 150 m block by the block (grid) contouring method using a 10 m × 10 m grid and a dumpy level/auto level.

Apparatus / Equipment Required

Auto level / dumpy level with tripod, levelling staff; 30 m tape, ranging rods, pegs (16 × 16 grid), field book, drawing sheet, french curves.

Theory

A contour is an imaginary line joining points of equal elevation. In the indirect (block) method the area is divided into squares by a grid, the RL of every grid point is obtained by levelling, and the contours are then interpolated between the grid points.

Interpolation is done on the assumption that the ground slope between two adjacent grid points is uniform: distance of a contour point from a grid point = (contour value – RL of that point) / (difference of RL between the two points) × grid interval.

Formulae Used

- $x = (h_c - h_1) / (h_2 - h_1) \times d$
- where x = distance of the contour point from grid point 1, h_c = contour RL, h_1, h_2 = RLs of adjacent grid points, d = grid interval (10 m)

Procedure

1. The 150 m × 150 m block was divided into a 10 m × 10 m grid (16 lines each way) and pegs were fixed at every node.
2. The level was set up at a central position and the RL of the first node was obtained by a back sight on a bench mark.
3. The RLs of all grid points were obtained by taking staff readings, shifting the instrument whenever necessary.
4. The grid RLs were entered in a block-level table and the contour interval was selected as 0.50 m.
5. Contour points on the grid lines were interpolated and joined by smooth curves.
6. The contour map was inked with the index contours thickened, and a north arrow, scale and legend were added.

Observation

Grid line (Northing)	0 m	10 m	20 m	30 m
0 m	100.20	100.35	100.50	100.70
10 m	100.30	100.45	100.62	100.85
20 m	100.42	100.58	100.76	101.00
30 m	100.55	100.72	100.90	101.15

Sample block-level table (part of the 16 × 16 grid). Contour interval = 0.50 m.

Calculation

Take grid points (10 m, 0 m) with RL 100.35 and (20 m, 0 m) with RL 100.50. For the 100.50 m contour:

x = (100.50 – 100.35) / (100.50 – 100.35) × 10 = **10.00 m**, i.e. the contour passes through the grid point (20 m, 0 m).

For grid points (0 m, 10 m) RL 100.30 and (10 m, 10 m) RL 100.45, the 100.40 m contour lies at x = (100.40 – 100.30)/(100.45 – 100.30) × 10 = **6.67 m** from the first point.

The slopes obtained from the grid show a general rise towards the north-east, with a maximum RL of 101.15 m at (30 m, 30 m) and a minimum of 100.20 m at the origin.

Result

A contour map of the 150 m × 150 m block was prepared with a contour interval of 0.50 m. The ground rises uniformly from the south-west corner (RL 100.20 m) towards the north-east, and the contours are nearly parallel and evenly spaced, indicating a uniform gentle slope.

Precautions

- The grid must be set out accurately with the tape; errors in the grid shift all contours.
- Take readings quickly to minimise changes in the level due to temperature.
- Interpolate contours only between adjacent grid points and never across a break in the slope.
- Draw contours as smooth flowing curves — they should never cross each other or branch.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 12

Measurement of area of an irregular figure using a digital planimeter

Aim

To measure the area of a given irregular figure (map/plan) with a digital planimeter and to convert it to the actual ground area using the map scale.

Apparatus / Equipment Required

Digital planimeter with tracer arm and roller; Given map/plan, drawing sheet fixed on a smooth board, pencil, scale.

Theory

A planimeter is a mechanical/digital integrating instrument used to measure the area of an irregular figure directly by tracing its boundary. When the tracer point is moved along the outline, the measuring wheel (or the digital counter) records the area in terms of an instrument constant.

Area = C × (final reading – initial reading) × M, where C is the planimeter constant (value of one unit of the counter) and M is the map multiplier, M = (scale of map)².

Formulae Used

- Area on map = C × (F.R. – I.R.)
- Actual area = Area on map × M, where M = (denominator of RF)²

Procedure

1. The plan was fixed flat on a smooth drawing board and the planimeter was placed near the figure.
2. The tracer point was set on a convenient starting point on the boundary and the initial reading was noted.
3. The tracer point was moved slowly and steadily along the boundary in a clockwise direction until it returned to the starting point.
4. The final reading was noted; the operation was repeated three times and the mean was taken.
5. The area on the map was computed as C × (F.R. – I.R.) and converted to the actual area using the map scale.
6. The result was verified by measuring the area of a known circle/square.

Observation

Trial	Initial reading (I.R.)	Final reading (F.R.)	Difference	Area on map (cm²)
1	2.436	8.452	6.016	60.16
2	2.438	8.449	6.011	60.11
3	2.441	8.455	6.014	60.14
Mean	—	—	6.014	60.14

Planimeter constant C = 10 cm² per unit reading; map scale 1:100.

Calculation

Mean difference = (6.016 + 6.011 + 6.014) / 3 = 6.014

Area on map = C × (F.R. – I.R.) = 10 × 6.014 = **60.14 cm²**

Map scale = 1:100, hence M = (100)² = 10,000 and 1 cm² on map = 1 m² on ground

Actual area = 60.14 × 1 = **60.14 m²**

Check: the known 50 m² plot gave a reading of 5.002 units → 50.02 m² (error 0.04%), hence the instrument constant is correct.

Result

The area of the given irregular figure is **60.14 cm²** on the map, which corresponds to an actual area of **60.14 m²** at a map scale of 1:100.

Precautions

- Fix the map firmly so that it does not shift while tracing.
- Move the tracer point slowly and exactly along the boundary; do not lift it during the run.
- Set the starting point on a straight portion of the boundary for better repeatability.
- Always check the planimeter against a known area before taking readings.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	B.C. Punmia, *Surveying* Vol. I, Laxmi Publications, New Delhi.
2	S.K. Duggal, *Surveying and Levelling*, McGraw Hill Education.
3	R. Agor, *A Text Book of Surveying and Levelling*, Khanna Publishers.
4	SBTE Bihar curriculum for Basic Surveying (2415301 / P2415301) — LSO 1.1 to 12.1.
5	Operating manual of the digital planimeter used in the laboratory.

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.

STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

STRENGTH OF MATERIAL FOR CIVIL ENGG.

Course Code: 2415303 (P2415303)

Diploma in Civil Engineering • Semester III

Student's Name	
Enrollment No.	
Roll No.	
Branch	Civil Engineering
Semester	III
Session	20____ – 20____
Name of Institute	

Department of Civil Engineering

(Name & address of the institute)

How to write this file

Is file ke har practical ko nimn likhit format mein likha gaya hai. Apne college ke lab manual se practical ki sankhya aur readings confirm karke observation table mein apni actual readings bharein — yahan di gayi readings sample hain, sirf samajhne ke liye.

- **Aim** — practical ka uddeshya ek line mein.
- **Apparatus** — naam aur specification (capacity, IS code).
- **Theory / Formula** — principle aur formula jo calculation mein use hua.
- **Procedure** — step-wise, past tense mein, chhote vakya.
- **Observation table** — readings, units ke saath; IS code ka reference.
- **Calculation** — formula lagakar steps dikhayein.
- **Result** — answer with unit aur IS permissible limit ke saath.
- **Precautions** — 3 se 4 points.

General instructions

- Har practical ke baad date aur teacher ke signature zaroor lein.
- Diagram pencil se, scale se banayein; likhai blue/black pen se.
- Index page mein page number bharna na bhoolein — file check ke waqt yahi dekha jata hai.
- Practical exam (ELA) mein viva hota hai — aim, formula aur result yaad rakhein.

Index

Ye file SBTE Bihar curriculum P2415303 ke 13 practicals par aadharit hai. Page number index mein bharein aur har practical par teacher ka signature lein.

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03	Tension test on mild steel bar as per IS: 1608	6		
04	Compression test on a mild steel specimen using Universal Testing Machine	7		
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06	Shear strength test on a given specimen	9		
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PRACTICAL NO. 01

Determination of moment of inertia of a flywheel and comparison with its theoretical value

Aim

To determine the moment of inertia of a given flywheel experimentally and to compare it with the theoretical value calculated from its mass and dimensions.

Apparatus / Equipment Required

Flywheel mounted on ball bearings with axle and pointer; Set of weights, strong thread, stopwatch, vernier caliper, steel scale, weighing balance.

Theory

The moment of inertia of a body about an axis is the sum of the products of the masses of its particles and the squares of their distances from that axis, $I = \Sigma mr^2$; it measures the resistance of the body to angular acceleration. In this experiment a weight m hangs from a thread wound round the axle of radius r, and when it falls through a height h in time t its potential energy is converted into the kinetic energy of the weight and the rotational kinetic energy of the flywheel, part of the energy being lost in friction.

Formulae Used

- Velocity of the falling weight, $v = 2h / t$
- Angular velocity of the flywheel, $\omega = v / r$
- Loss of potential energy = $m g h$; K.E. of weight = $\tfrac{1}{2} m v^2$; rotational K.E. of flywheel = $\tfrac{1}{2} I \omega^2$
- $I = [\, 2 m g h - m v^2 \,] / \omega^2$ (friction neglected)
- Theoretical value for a disc, $I = \tfrac{1}{2} M R^2$

Procedure

1. The mass M and radius R of the flywheel, and the radius r of the axle, were measured.
2. A thread carrying a weight of 2 kg was wound round the axle, the weight being kept at a known height h.
3. The weight was released and the time t taken to reach the ground was measured with a stopwatch.
4. The experiment was repeated five times with the same weight and the mean time was taken.
5. The velocity v, angular velocity ω and the moment of inertia I were calculated.
6. The experimental value was compared with the theoretical value $\tfrac{1}{2}MR^2$.

Observation

Trial	Height h (m)	Time t (s)	v = 2h/t (m/s)	ω = v/r (rad/s)
1	1.200	8.10	0.296	11.85
2	1.200	8.05	0.298	11.92
3	1.200	8.00	0.300	12.00
Mean	1.200	8.05	0.298	11.92

Mass of falling weight m = 2.0 kg; radius of axle r = 0.025 m; mass of flywheel M = 25.0 kg; radius R = 0.160 m. Take g = 9.81 m/s². Readings are sample values.

Calculation

$v = 2 \times 1.200 / 8.05 = 0.298$ m/s ; $\omega = 0.298 / 0.025 =$ **11.92 rad/s**

P.E. lost = $m g h = 23.54$ J; K.E. of weight = $\tfrac{1}{2} \times 2.0 \times 0.298^2 = 0.089$ J; hence rotational K.E. = $23.54 - 0.089 = 23.45$ J = $\tfrac{1}{2} I \omega^2$

$I = (2 \times 23.45) / (11.92)^2 = 46.90 / 142.09 =$ **0.330 kg·m²**

Theoretical $I = \tfrac{1}{2} M R^2 = 0.5 \times 25.0 \times 0.160^2 =$ **0.320 kg·m²**

Percentage difference = $(0.330 - 0.320)/0.320 \times 100 =$ **3.1 %**

Result

The moment of inertia of the flywheel determined experimentally is **0.330 kg·m²**, while its theoretical value is **0.320 kg·m²**. The difference is 3.1%, which is due to friction at the bearings and is within acceptable limits.

Precautions

- The thread must be wound uniformly on the axle and should not overlap.
- Start the stopwatch exactly when the weight is released and stop it when the weight touches the ground.
- Avoid parallax while measuring the height and radius.
- Take several trials and use the mean time to reduce random error.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 02

Study of different components of a Universal Testing Machine (UTM) and their functions

Aim

To study the construction and working of a Universal Testing Machine and to identify its different components with their functions.

Apparatus / Equipment Required

Universal Testing Machine (capacity 400 kN / 1000 kN) with accessories; Tension grips, compression platens, extensometer, dial gauge, measuring scale.

Theory

A Universal Testing Machine is used to conduct tension, compression, bending and shear tests on metals, timber and concrete. It consists of a load frame with a fixed crosshead and a movable crosshead, a hydraulic loading system and a load-measuring arrangement.

The load applied is measured either by a pendulum dynamometer or by a pressure gauge/load cell, and the deformation is measured by an extensometer or a dial gauge. From the load-deformation curve the yield strength, ultimate strength, modulus of elasticity and percentage elongation of the material can be determined.

Procedure

1. The machine was examined with the power supply switched off and the parts were identified.
2. The load frame, upper fixed crosshead and lower movable crosshead were studied.
3. The hydraulic pump, control valves, pressure gauge and load-indicating dial were identified.
4. The tension grips, compression platens, bending and shear attachments were examined.
5. The extensometer, dial gauge and the autographic recorder were studied along with their use.
6. The working of the machine — gripping, loading, rate control and unloading — was demonstrated.

Observation

S. No.	Component	Function
1	Base and columns	Support the frame and guide the movable crosshead
2	Fixed (upper) crosshead	Holds one end of the specimen through grips
3	Movable (lower) crosshead	Applies load to the specimen by hydraulic pressure
4	Hydraulic pump and valves	Generate and control the oil pressure for loading
5	Load measuring dial / load cell	Indicates the load applied on the specimen
6	Extensometer / dial gauge	Measures elongation or deformation of the specimen
7	Grips and platens	Hold the specimen for tension, compression, bending or shear tests
8	Autographic recorder	Plots the load-deformation curve automatically

Machine capacity: 400 kN; loading rate for steel test: 10 N/mm² per second.

Calculation

No calculation is required in this study. The relation between load and stress is:

Stress = Load / Cross-sectional area, and Strain = Change in length / Original length.

Modulus of elasticity E = Stress / Strain, obtained from the initial straight portion of the load-deformation curve.

Result

The different components of the Universal Testing Machine were studied and their functions were understood. The machine can perform tension, compression, bending and shear tests, and the load is measured by the pendulum dynamometer / load cell while deformation is measured by the extensometer.

Precautions

- Never operate the machine without the permission of the laboratory in charge.
- Ensure that the specimen is properly gripped and aligned before loading.
- Do not exceed the capacity of the machine.
- Keep hands away from the crossheads and grips during the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 03

Tension test on mild steel bar as per IS: 1608

Aim

To determine the yield strength, ultimate tensile strength, breaking strength and percentage elongation of a given mild steel bar by a tension test.

Apparatus / Equipment Required

Universal Testing Machine with tension grips; Mild steel specimen (12 mm diameter, gauge length 60 mm), punch, steel scale, vernier caliper.

Theory

When a ductile material such as mild steel is subjected to a gradually increasing axial tensile load it passes through the proportional limit, elastic limit, upper and lower yield points, strain hardening and ultimate load, and finally fractures. The yield stress is the load at which plastic deformation begins and the ultimate stress is the maximum load divided by the original area, while the percentage elongation of the gauge length after fracture indicates the ductility of the material.

Formulae Used

- Area, $A = \pi d^2 / 4$
- Yield stress = yield load / original area ; Ultimate stress = maximum load / original area
- Percentage elongation = [(L■ – L■) / L■] × 100
- Percentage reduction in area = [(A■ – A■) / A■] × 100

Procedure

1. The diameter of the specimen was measured at three places with a vernier caliper and the mean was taken.
2. A gauge length of 60 mm was marked on the specimen and the specimen was fitted in the grips of the UTM.
3. The load was applied gradually and the yield point, maximum load and breaking load were noted.
4. The load-deformation curve was recorded on the autographic recorder.
5. The two broken pieces were joined and the final gauge length was measured.
6. The stresses and percentage elongation were calculated.

Observation

Particular	Value	Unit
Original diameter / area	12.0 / 113.1	mm / mm²
Original gauge length, L■	60.0	mm
Yield load / maximum load	33.0 / 47.0	kN
Breaking load	36.5	kN
Final gauge length, L■	78.0	mm
Diameter at fracture	8.4	mm

Material: mild steel (Fe 415 grade); rate of loading 10 N/mm²/s. Readings are sample values.

Calculation

$A = \pi \times 12.0^2/4 =$ **113.1 mm²**

Yield stress = $33.0 \times 10^3 / 113.1 =$ **291.8 N/mm²**

Ultimate tensile stress = $47.0 \times 10^3 / 113.1 =$ **415.6 N/mm²**

Breaking stress = $36.5 \times 10^3 / 113.1 =$ **322.7 N/mm²**

Percentage elongation = (78.0 – 60.0)/60.0 × 100 = **30.0 %**; final area = 55.4 mm², hence reduction in area = (113.1 – 55.4)/113.1 × 100 = **51.0 %**

Result

For the given mild steel specimen, the yield stress is **291.8 N/mm²**, the ultimate tensile strength is **415.6 N/mm²**, the breaking stress is **322.7 N/mm²**, the percentage elongation is **30%** and the reduction in area is **51%**. The material is ductile and conforms to Fe 415 grade steel.

Precautions

- Measure the diameter accurately at three sections and use the mean value.
- Apply the load gradually and without shock, at the specified rate.
- Mark the gauge length before the test and measure the final length after joining the broken pieces.
- Note the yield point carefully — the pointer may pause or drop at the yield stage.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 04

Compression test on a mild steel specimen using Universal Testing Machine

Aim

To determine the compressive strength and behaviour of a given mild steel specimen under axial compression using a Universal Testing Machine.

Apparatus / Equipment Required

Universal Testing Machine with compression platens; Mild steel specimen (12 mm diameter, 24 mm length), vernier caliper, steel scale.

Theory

When a short specimen is subjected to axial compression it shortens and bulges laterally. For a ductile material such as mild steel, the specimen does not fracture but flattens, so the test gives the yield strength in compression rather than a breaking stress.

The compressive stress is the load divided by the original cross-sectional area. For ductile materials the compressive yield stress is nearly equal to the tensile yield stress, whereas brittle materials are much stronger in compression than in tension.

Formulae Used

- Area, $A = \pi d^2 / 4$
- Compressive stress = load / original area
- Percentage decrease in height = [(h■ – h■) / h■] × 100

Procedure

1. The diameter and height of the specimen were measured with a vernier caliper.
2. The specimen was placed centrally between the two compression platens of the UTM.
3. The load was applied gradually and the yield load was noted from the dial/autographic record.
4. Loading was continued until the specimen flattened to about half of its original height.
5. The final height was measured and the compressive stress was calculated.

Observation

Particular	Value	Unit
Original diameter	12.0	mm
Original height, h■	24.0	mm
Original area	113.1	mm²
Yield load (compression)	32.0	kN
Maximum load applied	58.0	kN
Final height, h■	14.0	mm

Material: mild steel; loading rate 10 N/mm²/s. Readings are sample values.

Calculation

$A = \pi \times 12^2/4 =$ **113.1 mm²**

Compressive yield stress = $32.0 \times 10^3 / 113.1 =$ **283.0 N/mm²**

Maximum compressive stress = $58.0 \times 10^3 / 113.1 =$ **512.8 N/mm²**

Percentage decrease in height = (24.0 – 14.0)/24.0 × 100 = **41.7 %**

The compressive yield stress (283 N/mm²) is nearly equal to the tensile yield stress of mild steel (about 292 N/mm²), which confirms the ductile behaviour of the material.

Result

The compressive yield stress of the given mild steel specimen is **283.0 N/mm²** and the specimen flattened by about 42% of its original height without fracture. The material behaves as a ductile material in compression.

Precautions

- Place the specimen exactly at the centre of the lower platen to avoid eccentric loading.
- Apply the load slowly and uniformly; avoid shock loading.
- Do not continue loading beyond the capacity of the machine or after heavy bulging.
- Measure the height and diameter before and after the test with the same instrument.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 05

Flexural test on a timber beam in both orientations as per IS: 2408

Aim

To determine the flexural strength (modulus of rupture) of a given timber beam by testing it in two orientations of the rectangular section.

Apparatus / Equipment Required

Universal Testing Machine / flexure testing arrangement with two supports and a loading nose; Timber specimen (50 mm × 75 mm × 750 mm), vernier caliper, steel scale, dial gauge.

Theory

When a simply supported beam is loaded at the centre, bending stresses are set up in the section. The maximum bending stress at failure, called the modulus of rupture, is given by the flexure formula f = M/Z, where M is the maximum bending moment and Z the section modulus.

For a rectangular section of width b and depth d, Z = bd²/6. The flexural strength therefore depends on the orientation of the section: a beam is much stronger when loaded on its narrower face, because the depth is then greater.

Formulae Used

- Maximum bending moment, M = W L / 4 (central point load)
- Section modulus, Z = b d² / 6
- Modulus of rupture, f = M / Z = 3 W L / (2 b d²)

Procedure

1. The dimensions of the timber specimen were measured accurately.
2. The specimen was placed on the two supports of the testing arrangement with a span of 600 mm, on its wider face (orientation 1).
3. The load was applied at the centre through the loading nose at a uniform rate.
4. The maximum load at failure was noted and the modulus of rupture calculated.
5. The test was repeated on an identical specimen placed on its narrower face (orientation 2).
6. The two values were compared.

Observation

Orientation	b (mm)	d (mm)	Span L (mm)	Failure load W (kN)
1 — on wider face	50	75	600	3.20
2 — on narrower face	75	50	600	1.42

Timber: teak; moisture content about 12%. Readings are sample values.

Calculation

Orientation 1: f = 3 W L/(2 b d²) = (3 × 3200 × 600)/(2 × 50 × 75²) = 5,760,000/562,500 = **10.24 N/mm²**

Orientation 2: f = (3 × 1420 × 600)/(2 × 75 × 50²) = 2,556,000/375,000 = **6.82 N/mm²**

Ratio of strengths = 10.24/6.82 = 1.50, which equals the ratio of the depths (75/50 = 1.5), showing that the flexural strength is proportional to the depth of the section.

Result

The modulus of rupture of the given timber is **10.24 N/mm²** when the beam is loaded on its wider face and **6.82 N/mm²** when loaded on its narrower face. The strength in the first orientation is 1.5 times that in the second, confirming that the flexural strength is proportional to the square of the depth.

Precautions

- The specimen must be free from knots, cracks and other visible defects.
- Apply the load exactly at the centre of the span and without impact.
- Measure the dimensions at the section of failure.
- Keep the supports and loading nose parallel and firmly in contact with the specimen.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 06

Shear strength test on a given specimen

Aim

To determine the ultimate shear strength of a given mild steel specimen by a single shear test.

Apparatus / Equipment Required

Universal Testing Machine with shear test attachment (shear block and punch); Mild steel specimen (12 mm diameter), vernier caliper, steel scale.

Theory

When a specimen is subjected to a force acting parallel to its cross-section, shear stress is developed. In a single shear test the specimen is held in a shear block and a punch shears it across one plane, so the resisting area is the cross-sectional area of the specimen.

The shear strength is the maximum load divided by the sheared area. For mild steel the ultimate shear strength is approximately 0.5 to 0.6 times the ultimate tensile strength.

Formulae Used

- Sheared area, $A = \pi d^2 / 4$
- Ultimate shear strength, τ = shear load / sheared area

Procedure

1. The diameter of the specimen was measured and its cross-sectional area calculated.
2. The specimen was inserted in the shear block of the attachment, which was placed in the UTM.
3. The load was applied gradually until the specimen was sheared off.
4. The maximum shear load was noted from the load indicating dial.
5. The shear strength was calculated and compared with the tensile strength of the material.

Observation

Particular	Value	Unit
Diameter of specimen	12.0	mm
Sheared area	113.1	mm²
Shear load at failure	24.0	kN
Type of shear	Single shear	—

Material: mild steel. Readings are sample values.

Calculation

$A = \pi \times 12^2/4 =$ **113.1 mm²**

$\tau = 24.0 \times 10^3 / 113.1 =$ **212.2 N/mm²**

Compare with tensile strength of mild steel (≈ 415 N/mm²): $\tau/f_u = 212.2/415.6 = 0.51$, which agrees with the usual relation $\tau \approx 0.5\ f_u$.

Result

The ultimate shear strength of the given mild steel specimen is **212.2 N/mm²**, which is about 51% of its ultimate tensile strength, as expected for ductile steel.

Precautions

- Place the specimen exactly in the centre of the shear block to avoid bending.
- Tighten the attachment firmly so that the specimen does not slip.
- Apply the load slowly and uniformly until shear occurs.
- Measure the diameter at the sheared section after the test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 07

Izod impact test on a given specimen

Aim

To determine the impact strength (toughness) of a given metal specimen by the Izod impact test.

Apparatus / Equipment Required

Izod impact testing machine with a 167 J / 300 J pendulum hammer; Izod test specimen with a V-notch (10 mm × 10 mm section, 75 mm length), vernier caliper.

Theory

Toughness is the ability of a material to absorb energy before fracture. The Izod test measures this energy by breaking a notched specimen with a swinging pendulum; the specimen is held as a cantilever and struck at the notch.

The energy absorbed by the specimen is the difference between the energy of the hammer before and after fracture, and the impact strength is expressed as the energy absorbed per unit area at the notch.

Formulae Used

- Energy absorbed, E = W (h■ – h■) (J)
- Impact strength = energy absorbed / area of the notch (J/mm²)
- Impact value = energy absorbed by specimen / energy of the hammer × 100

Procedure

1. The dimensions of the specimen and the notch were measured.
2. The machine was checked with the pointer at the zero position and the pendulum was raised to its starting position.
3. The specimen was clamped in the vice with the notch facing the striker and the pointer was set.
4. The pendulum was released and the energy absorbed by the specimen was read from the scale.
5. The test was repeated on two more specimens and the mean impact strength was calculated.

Observation

Specimen	Energy absorbed (J)	Notch area (mm²)	Impact strength (J/mm²)
1	12.0	80	0.150
2	11.5	80	0.144
3	12.5	80	0.156
Mean	12.0	80	0.150

Material: mild steel; notch 45° V-notch, 2 mm deep. Readings are sample values.

Calculation

Mean energy absorbed = (12.0 + 11.5 + 12.5)/3 = **12.0 J**

Notch area = 10 mm × (2 × 4 mm) = 80 mm² (approximate fractured area)

Impact strength = 12.0 / 80 = **0.150 J/mm²**

Impact value = (12.0 / 167) × 100 = **7.2 %** of the hammer energy

Result

The Izod impact strength of the given mild steel specimen is **12.0 J** (0.150 J/mm²), which indicates good toughness and ductility of the material.

Precautions

- The pendulum must be locked in the raised position before the specimen is fixed.
- Keep clear of the swing path of the pendulum during the test.
- Locate the specimen accurately in the vice with the notch exactly opposite the striker.
- Do not touch the specimen or the machine before the pendulum comes to rest.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 08

Charpy impact test on a metal specimen

Aim

To determine the impact energy absorbed by a given metal specimen using the Charpy impact test.

Apparatus / Equipment Required

Charpy impact testing machine with pendulum hammer and scale; Charpy V-notch specimen (10 mm × 10 mm × 55 mm), vernier caliper, setting gauge.

Theory

In the Charpy test the notched specimen is supported as a simple beam at both ends and is struck by the pendulum at the centre, on the face opposite the notch. Because the specimen is supported at both ends, the Charpy test gives a slightly different measure of toughness than the Izod test.

The energy absorbed in breaking the specimen is read directly from the graduated scale of the machine and is a measure of the resistance of the material to sudden loading.

Formulae Used

- Energy absorbed, $E = W(\cos\beta - \cos\alpha)$ (J)
- Impact strength = energy absorbed / cross-sectional area at the notch (J/mm²)

Procedure

1. The specimen was prepared with a 45° V-notch of 2 mm depth and its dimensions were measured.
2. The pendulum was raised to the starting angle and the pointer was set to zero.
3. The specimen was placed centrally on the supports with the notch away from the striker.
4. The pendulum was released and the energy absorbed was read from the scale after the swing.
5. The test was repeated on two more specimens and the mean value was taken.

Observation

Specimen	Energy absorbed (J)	Area at notch (mm²)	Impact strength (J/mm²)
1	15.0	80	0.188
2	14.5	80	0.181
3	15.5	80	0.194
Mean	15.0	80	0.188

Material: mild steel; Charpy V-notch specimen. Readings are sample values.

Calculation

Mean energy absorbed = (15.0 + 14.5 + 15.5)/3 = **15.0 J**

Impact strength = 15.0 / 80 = **0.188 J/mm²**

The Charpy value (15.0 J) is slightly higher than the Izod value (12.0 J) of the same material because the specimen is supported at both ends and the test geometry is different.

Result

The Charpy impact energy absorbed by the given mild steel specimen is **15.0 J** (0.188 J/mm²), indicating good impact toughness of the material.

Precautions

- Place the specimen exactly on the supports using the setting gauge.
- Ensure that the notch is on the side away from the striker.
- Release the pendulum smoothly from the correct height.
- Read the scale after the pendulum has completed its forward swing.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 09

Brinell hardness test on a metal specimen

Aim

To determine the Brinell hardness number (BHN) of a given metal specimen.

Apparatus / Equipment Required

Brinell hardness testing machine with a hardened steel ball indenter (10 mm); Microscope or Brinell index measuring scale, specimen, standard test block.

Theory

Hardness is the resistance of a material to indentation. In the Brinell test a hardened steel ball of diameter D is pressed into the surface of the specimen under a standard load P and the diameter d of the resulting indentation is measured after removing the load.

The Brinell hardness number is the load divided by the curved surface area of the indentation, and is expressed in kg/mm² (or in MPa, numerically equal to BHN × 9.81).

Formulae Used

- $BHN = 2P / [\pi D (D - \sqrt{(D^2 - d^2)})]$
- where P = load in kg, D = diameter of ball in mm and d = diameter of indentation in mm
- Load $P = 30 D^2$ for steel (i.e. 3000 kg for a 10 mm ball)

Procedure

1. The surface of the specimen was ground smooth and cleaned.
2. The specimen was placed on the anvil of the machine and the 10 mm ball indenter was brought into contact with the surface.
3. A load of 3000 kg was applied and maintained for 15 seconds.
4. The load was released, the specimen was removed and the diameter of the indentation was measured in two perpendicular directions with the microscope.
5. The mean diameter was used to calculate the Brinell hardness number.

Observation

Test	Load P (kg)	Ball dia D (mm)	Indentation dia d (mm)	BHN
1	3000	10	4.20	207
2	3000	10	4.25	202
3	3000	10	4.15	212
Mean	3000	10	4.20	207

Material: mild steel; dwell time 15 s. Readings are sample values.

Calculation

For d = 4.20 mm: $\sqrt{(D^2 - d^2)} = \sqrt{(100 - 17.64)} = \sqrt{82.36} = 9.075$ mm

$BHN = 2 \times 3000 / [\pi \times 10 \times (10 - 9.075)] = 6000 / (31.416 \times 0.925) = 6000/29.06 =$ **206.5**

Mean BHN of the three tests = (207 + 202 + 212)/3 ≈ **207**

Result

The Brinell hardness number of the given mild steel specimen is **207** (≈ 2030 N/mm²).

Precautions

- The test surface must be clean, smooth and flat, and free from scale.
- Keep the indentation away from the edge of the specimen (at least 2.5 d).
- Apply the load slowly and maintain it for the specified dwell time.
- Measure the indentation diameter in two perpendicular directions and use the mean.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 10

Rockwell hardness test on a metal specimen

Aim

To determine the Rockwell hardness (HRC) of a given hardened steel specimen.

Apparatus / Equipment Required

Rockwell hardness testing machine with diamond cone indenter (120°); Specimen, standard test block, anvil.

Theory

In the Rockwell test the depth of indentation is measured directly by the machine instead of the diameter, which makes the test very quick. A minor load of 10 kg is first applied to seat the indenter, and then a major load of 150 kg is applied; the Rockwell hardness is read on the dial after the major load is removed.

The Rockwell C scale (HRC) uses a diamond cone indenter with a total load of 150 kg and is suitable for hardened steel. Higher hardness gives a higher HRC value, and each division of the dial corresponds to 0.002 mm of indentation depth.

Formulae Used

- HRC = 100 – (depth of indentation in divisions × 0.002 mm)
- The hardness is read directly on the dial gauge of the machine.

Procedure

1. The specimen surface was cleaned and polished, and the specimen was placed on the anvil.
2. The diamond cone indenter was brought into contact with the surface and the minor load of 10 kg was applied.
3. The dial was set to zero and the major load of 150 kg was applied for 10–15 seconds.
4. The major load was removed and the hardness was read on the C scale of the dial.
5. Five readings were taken at different points and the mean HRC was calculated.

Observation

Reading	1	2	3	4	5	Mean
HRC value	42	41	43	42	41	41.8

Material: hardened steel; indenter: diamond cone 120°; total load 150 kg (Scale C).

Calculation

Mean HRC = (42 + 41 + 43 + 42 + 41)/5 = **41.8**

Approximate tensile strength from HRC ≈ 3.5 × HRC = 3.5 × 41.8 ≈ **146 kg/mm²** (≈ 1435 N/mm²), which is typical of hardened and tempered steel.

Result

The Rockwell hardness of the given hardened steel specimen is **41.8 HRC**, which indicates that the material is hard and suitable for tools and machine parts.

Precautions

- Test on a smooth, clean surface; roughness gives erroneous readings.
- Keep successive indentations at least 3 mm apart and away from the edges.
- Apply the minor load first and set the dial to zero before applying the major load.
- Clean the indenter and anvil after every test.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 11

Preparation of models of different types of trusses

Aim

To prepare models of different types of trusses and to distinguish between them on the basis of their geometry and member arrangement.

Apparatus / Equipment Required

Balsa wood / match sticks / straw, glue, thread, pins; Drawing sheet, scale, protractor, cutting blade, sketch pens.

Theory

A truss is a framework of straight members connected at their ends by pin joints so that the members carry mainly axial forces. A truss is called perfect if the number of members m and the number of joints j satisfy $m = 2j - 3$; if $m < 2j - 3$ the truss is deficient (unstable), and if $m > 2j - 3$ it is redundant.

Common types of trusses are the king post truss (span up to 6 m), the queen post truss (6–9 m), the howe and pratt trusses, the warren truss, the fink truss and the north-light roof truss. The choice depends on the span, the load and the material used.

Formulae Used

- Degree of indeterminacy for a plane truss, $m = 2j - 3$ (perfect truss)
- where m = number of members and j = number of joints

Procedure

1. The required number of balsa wood pieces was cut to the calculated lengths of the members.
2. The joints were marked on a drawing sheet and the members were assembled as per the geometry of the truss.
3. The joints were glued and pinned and the model was allowed to dry.
4. Three models — king post, queen post and warren truss — were prepared in the same manner.
5. For every model, the number of members and joints were counted and the condition $m = 2j - 3$ was checked.
6. The models were labelled and compared with sketches.

Observation

Type of truss	Joints (j)	Members (m)	2j – 3	Remarks
King post	6	9	9	Perfect truss
Queen post	8	13	13	Perfect truss
Warren truss	10	17	17	Perfect truss
Deficient truss (without one diagonal)	6	8	9	Deficient — unstable

Models prepared to a scale of 1:50. Joints were pinned and glued.

Calculation

King post: $m = 2 \times 6 - 3 = 9$; actual members = 9 ✓ (perfect truss)

Queen post: $m = 2 \times 8 - 3 = 13$; actual members = 13 ✓ (perfect truss)

Warren: $m = 2 \times 10 - 3 = 17$; actual members = 17 ✓ (perfect truss)

Model with one member removed: $m = 8 < 9$, hence the truss becomes deficient and can collapse under load.

Result

Models of the king post, queen post and warren trusses were prepared successfully, and all of them satisfy the condition $m = 2j - 3$ for a perfect truss. The model with one member removed was found to be deficient, confirming the need for the correct number of members.

Precautions

- Cut the members to accurate lengths, otherwise the joints will not close properly.
- Keep the joints exactly at the marked points to maintain the geometry of the truss.
- Allow sufficient time for the adhesive to set before handling the model.
- Handle the models carefully while shifting and displaying them.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 12

Shear force and bending moment diagrams for a simply supported beam under point and distributed loads

Aim

To draw the shear force and bending moment diagrams for a simply supported beam carrying a point load and a uniformly distributed load, and to locate the point of maximum bending moment.

Apparatus / Equipment Required

Drawing sheet, scale, pencil, French curves, calculator; Data of the beam: span, loads and positions.

Theory

Shear force at a section is the algebraic sum of all vertical forces on one side of the section, and bending moment is the algebraic sum of the moments of those forces about the section. The relation between load, shear force and bending moment is dM/dx = V and dV/dx = w.

For a simply supported beam the bending moment is zero at both ends, and the maximum bending moment occurs where the shear force changes sign (point of zero shear).

Formulae Used

- Reactions: $\Sigma M_A = 0$ and $\Sigma V = 0$
- Shear force V_x = sum of upward forces on the left of the section
- Bending moment $M_x = \Sigma$ (force × lever arm)
- Maximum BM occurs at the section where V = 0

Procedure

1. The beam of span 6 m carrying a point load of 10 kN at 2 m from the left support and a UDL of 2 kN/m over the whole span was considered.
2. The reactions at the supports were calculated using the equations of equilibrium.
3. Shear force was calculated at every critical section (supports, under the point load and points of zero shear).
4. Bending moment was calculated at the same sections.
5. The shear force and bending moment diagrams were drawn to a suitable scale below the beam sketch.

Observation

Section (m from A)	Shear force (kN)	Bending moment (kNm)	Remarks
0 (A)	+12.67	0	Left support
2.0 (just left of load)	+8.67	+25.33	Under 10 kN load
2.0 (just right of load)	–1.33	+25.33	Sign change
2.67	0	+25.79	Maximum BM (V = 0)
6.0 (B)	–9.33	0	Right support

Span 6 m; point load 10 kN at 2 m from A; UDL 2 kN/m over the whole span.

Calculation

Taking moments about A: $R_B \times 6 = 10 \times 2 + (2 \times 6) \times 3 = 20 + 36 = 56 \rightarrow R_B =$ **9.33 kN**

R_A = total load – R_B = (10 + 12) – 9.33 = **12.67 kN**

SF at A = +12.67 kN; just left of the load = 12.67 – 2 × 2 = +8.67 kN; just right = 8.67 – 10 = –1.33 kN

Point of zero shear: 12.67 – 2x – 10 = 0 → x = **1.33 m** from the load, i.e. 2.67 m from A

M at 2 m = 12.67 × 2 – 2 × 2²/2 = 25.33 – 4.00 = **21.33 kNm** (left of load)

M_{max} at x = 2.67 m = 12.67 × 2.67 – 2 × 2.67²/2 – 10 × 0.67 = 33.83 – 7.13 – 6.70 = **20.00 kNm**

Check: M at B = 12.67 × 6 – 10 × 4 – 2 × 6²/2 = 76.02 – 40 – 36 = 0.02 ≈ **0** ✓

Result

The reactions are R_A = 12.67 kN and R_B = 9.33 kN. The maximum bending moment is **20.00 kNm** at 2.67 m from the left support, where the shear force changes sign, and the bending moment is zero at both ends.

Precautions

- Follow a proper sign convention throughout (upward force = positive shear).
- Calculate the reactions correctly before drawing the diagrams.
- Mark the points of zero shear and maximum bending moment clearly.
- Draw the diagrams to a stated scale and label the values at all critical sections.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 13

Determination of principal stresses and their plane using Mohr's circle

Aim

To determine the principal stresses, maximum shear stress and the position of the principal planes for a given state of stress using Mohr's circle.

Apparatus / Equipment Required

Drawing sheet, compass, protractor, scale, pencil, calculator; Given state of stress: σ_x = 80 N/mm², σ_y = 20 N/mm², τ = 30 N/mm².

Theory

At a point in a stressed body there are always two mutually perpendicular planes on which the shear stress is zero; the normal stresses on these planes are called principal stresses. Their values can be found analytically or graphically by Mohr's circle.

In Mohr's circle the normal stresses are plotted on the horizontal axis and the shear stresses on the vertical axis. The circle is drawn with the centre at $((\sigma_x + \sigma_y)/2, 0)$ and radius equal to the maximum shear stress; the points where the circle cuts the horizontal axis give the principal stresses.

Formulae Used

- Centre of circle, $C = (\sigma_x + \sigma_y) / 2$
- Radius, $R = \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau^2\,]}$
- $\sigma_1 = C + R$; $\sigma_2 = C - R$; $\tau_{max} = R$
- $\tan 2\theta = 2\tau / (\sigma_x - \sigma_y)$

Procedure

1. The given values of σ_x, σ_y and τ were noted.
2. The centre C and radius R of Mohr's circle were calculated.
3. The circle was drawn to a suitable scale with C as centre on the σ-axis.
4. The points of intersection of the circle with the σ-axis were marked as σ_1 and σ_2.
5. The maximum shear stress was measured as the radius and the angle 2θ was measured from the circle.
6. The values were compared with those obtained analytically.

Observation

Quantity	Graphical value	Analytical value	Unit
Centre of circle	50.0	50.0	N/mm²
Radius (τ_{max})	42.4	42.43	N/mm²
Major principal stress σ_1	92.4	92.43	N/mm²
Minor principal stress σ_2	7.6	7.57	N/mm²
Angle of principal plane θ	22.5°	22.5°	degree

Scale: 1 cm = 10 N/mm². σ_x = 80, σ_y = 20, τ = 30 N/mm².

Calculation

$C = (80 + 20)/2 =$ **50 N/mm²**

$R = \sqrt{[\,((80 - 20)/2)^2 + 30^2\,]} = \sqrt{(900 + 900)} = \sqrt{1800} =$ **42.43 N/mm²**

$\sigma_1 = 50 + 42.43 =$ **92.43 N/mm²**; $\sigma_2 = 50 - 42.43 =$ **7.57 N/mm²**

$\tau_{max} = R =$ **42.43 N/mm²**

$\tan 2\theta = (2 \times 30)/(80 - 20) = 60/60 = 1 \rightarrow 2\theta = 45° \rightarrow \theta =$ **22.5°**

Result

For the given state of stress the principal stresses are σ_1 **= 92.43 N/mm²** and σ_2 **= 7.57 N/mm²**, the maximum shear stress is **42.43 N/mm²**, and the principal planes are inclined at **22.5°** to the plane of σ_x. The graphical and analytical values agree closely.

Precautions

- Choose a suitable scale so that the circle fits on the sheet.
- Measure the angles from the correct reference axis (2θ on Mohr's circle).
- Plot the shear stress with the correct sign convention.
- Verify the graphical results with the analytical formulae.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	R.K. Bansal, *Strength of Materials*, Laxmi Publications.
2	B.C. Punmia, *Strength of Materials and Mechanics of Structures*, Laxmi Publications.
3	S. Ramamrutham, *Strength of Materials*, Dhanpat Rai Publishing Co.
4	IS: 1608 — tensile testing; IS: 2408 — timber; IS: 1977 — Izod impact.
5	IS: 1757 — Charpy impact; IS: 2281 — Brinell hardness; IS: 1586 (Part 2) — Rockwell hardness.
6	SBTE Bihar curriculum for Strength of Material (2415303 / P2415303).

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.

STATE BOARD OF TECHNICAL EDUCATION, BIHAR

LABORATORY

PRACTICAL FILE

BUILDING CONSTRUCTION & MATERIAL

Course Code: 2415304 (P2415304)

Diploma in Civil Engineering • Semester III

Student's Name	
Enrollment No.	
Roll No.	
Branch	Civil Engineering
Semester	III
Session	20____ – 20____
Name of Institute	

Department of Civil Engineering

(Name & address of the institute)

How to write this file

Is file ke har practical ko nimn likhit format mein likha gaya hai. Apne college ke lab manual se practical ki sankhya aur readings confirm karke observation table mein apni actual readings bharein — yahan di gayi readings sample hain, sirf samajhne ke liye.

- **Aim** — practical ka uddeshya ek line mein.
- **Apparatus** — naam aur specification (capacity, IS code).
- **Theory / Formula** — principle aur formula jo calculation mein use hua.
- **Procedure** — step-wise, past tense mein, chhote vakya.
- **Observation table** — readings, units ke saath; IS code ka reference.
- **Calculation** — formula lagakar steps dikhayein.
- **Result** — answer with unit aur IS permissible limit ke saath.
- **Precautions** — 3 se 4 points.

General instructions

- Har practical ke baad date aur teacher ke signature zaroor lein.
- Diagram pencil se, scale se banayein; likhai blue/black pen se.
- Index page mein page number bharna na bhoolein — file check ke waqt yahi dekha jata hai.
- Practical exam (ELA) mein viva hota hai — aim, formula aur result yaad rakhein.

Index

Ye file SBTE Bihar curriculum P2415304 ke practicals par aadharit hai (approx. 12 practicals). Apne college ke lab manual se practical ki sankhya confirm karein aur index mein page number bharein.

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PRACTICAL NO. 01

Identification and classification of building stones by field tests

Aim

To identify the given samples of building stones and to classify them on the basis of their physical properties and field tests.

Apparatus / Equipment Required

Stone samples (granite, sandstone, limestone, marble, laterite, slate); Geologist's hammer, magnifying glass, pen knife, weighing balance, measuring cylinder, dilute HCl.

Theory

Stone is one of the oldest building materials and is used for masonry, flooring, cladding and foundations. Stones are classified geologically into igneous (granite, basalt), sedimentary (sandstone, limestone) and metamorphic (marble, slate) rocks.

A good building stone should be hard, durable, compact, free from cracks and harmful salts, and should have sufficient strength and low water absorption. Field tests such as appearance, hardness, acid test and water absorption give a quick indication of its quality.

Procedure

1. The colour, texture and structure of each stone sample were observed with the naked eye and a magnifying glass.
2. The hardness of each sample was tested by scratching with a pen knife and observing the mark left.
3. A few drops of dilute hydrochloric acid were placed on each sample and the reaction was noted (effervescence indicates lime content).
4. The specific gravity and water absorption of each sample were determined by weighing the dry and saturated samples.
5. The samples were classified as igneous, sedimentary or metamorphic, and their suitability for different construction uses was noted.

Observation

Stone	Colour / texture	Hardness (knife)	Acid test	Class
Granite	Grey, crystalline, granular	No scratch	No reaction	Igneous
Sandstone	Reddish, granular	Slight scratch	No reaction	Sedimentary
Limestone	Cream, fine grained	Scratches easily	Effervescence	Sedimentary
Marble	White, crystalline	Scratches	Effervescence	Metamorphic
Laterite	Brown, porous	Scratches easily	Slight reaction	Sedimentary

Acid used: dilute HCl (1:10). Observations are sample values.

Calculation

Water absorption of granite = $[(W_{sat} - W_{dry})/W_{dry}] \times 100 = [(1030 - 1000)/1000] \times 100 =$ **3.0 %**

Water absorption of sandstone = $[(1120 - 1000)/1000] \times 100 =$ **12.0 %**

Lower water absorption indicates a more compact and durable stone; granite and marble are therefore more suitable for flooring and facing work, while sandstone and limestone are used for masonry.

Result

The given stone samples were identified as granite (igneous), sandstone and limestone (sedimentary), marble (metamorphic) and laterite. Granite was found to be the hardest and least porous, and is suitable for flooring and heavy masonry, whereas limestone and sandstone are suitable for general masonry work.

Precautions

- Handle the acid carefully and keep it away from the eyes and skin.
- Test hardness on a fresh, clean surface of the stone.
- Use a representative sample and avoid broken or weathered pieces.
- Record the observations immediately to avoid mixing up the samples.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 02

Field tests on bricks — shape, size, soundness, hardness and structure

Aim

To carry out the field tests on the given samples of bricks to judge their quality.

Apparatus / Equipment Required

Brick samples (10 nos.), steel scale, measuring tape, pen knife, weighing balance, two bricks.

Theory

Bricks are the most commonly used masonry units. A first class brick should have a uniform shape and standard size, sharp edges, a smooth surface, a clear metallic ringing sound when struck, and should not break when dropped from a height of about 1 m.

Field tests are quick tests carried out at the site to check the quality of bricks before they are used. Laboratory tests such as water absorption and compressive strength are carried out for confirming the quality.

Procedure

1. Ten bricks were taken at random from the stack and their dimensions were measured with a steel scale.
2. Each brick was examined for shape, edges, cracks, warping and uniformity of colour.
3. Two bricks were struck against each other and the ringing sound was noted.
4. A scratch was made on the surface of each brick with a pen knife to check its hardness.
5. One brick was dropped flat on hard ground from a height of about 1 m and the result was noted.
6. A broken brick was examined for a uniform, compact and free-from-lumps structure.

Observation

Test	Observation	Inference
Dimensions	230 × 110 × 70 mm (mean)	Standard modular size
Shape and edges	Sharp, square edges, uniform	Good
Colour	Uniform deep red	Well burnt
Sound test	Clear ringing sound	Good
Hardness test	No impression by finger nail; slight knife scratch	Good
Structure	Compact, no lumps or stones	Good
Drop test (1 m)	Did not break	Good

Bricks: conventional burnt clay bricks from a local kiln. Observations are sample values.

Calculation

Standard modular brick size = 190 × 90 × 90 mm (with mortar) and conventional size = 230 × 110 × 70 mm.

Percentage variation in length = (232 – 230)/230 × 100 = **0.87 %**, which is within the permissible tolerance of ±3%.

All the field tests indicate that the bricks are well burnt and of good quality (first class).

Result

The given bricks satisfy all the field tests — standard size, uniform colour, clear ringing sound, good hardness, compact structure and no breakage on dropping. Hence the bricks are of **first class quality** and suitable for masonry work.

Precautions

- Take samples at random from different parts of the stack.
- Measure the dimensions of at least ten bricks and use the mean value.
- Do not use broken or over-burnt bricks for the tests.
- Carry out the drop test on a hard, level surface.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 03

Determination of water absorption of bricks (IS: 3495 Part 2)

Aim

To determine the water absorption of the given bricks after 24 hours of immersion in cold water.

Apparatus / Equipment Required

Brick samples (3 nos.), weighing balance (accuracy 1 g), water tank, oven, absorbent cloth.

Theory

Water absorption of a brick indicates its porosity and, indirectly, its durability. A highly porous brick absorbs more water, which may cause efflorescence, dampness and disintegration on freezing.

As per IS: 1077, the water absorption of first class bricks shall not exceed 20% by mass, and for second class bricks it shall not exceed 22%.

Formulae Used

- Water absorption (%) = [(W■ – W■) / W■] × 100
- where W■ = mass of the oven-dry brick and W■ = mass of the brick after 24 hours of immersion

Procedure

1. The bricks were dried in an oven at 105–115 °C until constant mass and then cooled.
2. The dry mass W■ of each brick was recorded.
3. The bricks were immersed in clean water at 27 ± 2 °C for 24 hours, being kept fully covered.
4. The bricks were taken out, the surface water was wiped off with a damp cloth and the mass W■ was recorded.
5. The water absorption of each brick was calculated and the mean value reported.

Observation

Brick	Dry mass W■ (g)	Wet mass W■ (g)	Absorption (%)
1	3000	3360	12.0
2	2980	3338	12.0
3	3020	3382	12.0
Mean	3000	3360	12.0

Bricks: first class burnt clay bricks. Readings are sample values.

Calculation

Absorption of brick 1 = (3360 – 3000)/3000 × 100 = **12.0 %**

Absorption of brick 2 = (3338 – 2980)/2980 × 100 = **12.0 %**

Absorption of brick 3 = (3382 – 3020)/3020 × 100 = **12.0 %**

The mean absorption of 12% is less than the 20% limit specified for first class bricks.

Result

The water absorption of the given bricks is **12.0 %**, which is less than the permissible limit of 20% for first class bricks. Hence the bricks are sufficiently dense and durable for masonry work.

Precautions

- Dry the bricks to constant mass before the test.
- Keep the bricks fully immersed and separate from each other during soaking.
- Wipe off only the surface water before weighing — do not press the brick.
- Weigh the bricks immediately after wiping to avoid evaporation loss.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 04

Determination of compressive strength of bricks / blocks (IS: 3495 Part 1)

Aim

To determine the compressive strength of the given bricks (or blocks) by testing them in a compression testing machine.

Apparatus / Equipment Required

Compression testing machine (2000 kN), brick samples (5 nos.); Cement mortar (1:3) for capping, glass plate, steel scale, water bath, oven.

Theory

The compressive strength of a brick is the most important criterion of its quality, because masonry is mainly subjected to compressive loads. The test is carried out on bricks whose frog and bed faces are made plane and parallel by cement mortar capping.

As per IS: 1077, the minimum average compressive strength of common burnt clay bricks is 3.5 N/mm² for class 3.5, and a first class brick should have a strength of not less than 10 N/mm².

Formulae Used

- Compressive strength = maximum load at failure / area of the brick (N/mm²)
- Area = length × width of the brick

Procedure

1. Five bricks were taken and their dimensions were measured; the frog and bed faces were made plane.
2. The bricks were soaked in water for 24 hours before testing.
3. The bricks were removed from water and the surface water was wiped off.
4. Cement mortar (1:3) was applied on the bed faces to make them parallel and the bricks were kept under a moist cloth for 24 hours.
5. Each brick was placed centrally in the compression testing machine with the mortar-capped faces in contact with the platens.
6. The load was applied uniformly at the rate of 14 N/mm² per minute until failure, and the maximum load was recorded.

Observation

Brick	Length × width (mm)	Area (mm²)	Failure load (kN)	Strength (N/mm²)
1	230 × 110	25300	205	8.10
2	230 × 110	25300	215	8.50
3	230 × 110	25300	210	8.30
4	230 × 110	25300	220	8.70
5	230 × 110	25300	212	8.38

Bricks: burnt clay bricks; rate of loading 14 N/mm² per minute. Readings are sample values.

Calculation

Area = 230 × 110 = **25300 mm²**

Strength of brick 1 = 205 × 10^3 / 25300 = **8.10 N/mm²**

Mean strength = (8.10 + 8.50 + 8.30 + 8.70 + 8.38)/5 = **8.40 N/mm²**

As the mean strength is between 7.5 and 10 N/mm², the bricks conform to **class 7.5** of IS: 1077 (second class bricks), not to first class.

Result

The average compressive strength of the given bricks is **8.40 N/mm²**, hence they conform to **class 7.5** of IS: 1077 and can be used for load-bearing masonry with a suitable factor of safety.

Precautions

- The bed faces must be plane and parallel; capping with cement mortar is essential.
- Soak the bricks before testing to avoid absorption of water from the mortar.
- Place the brick centrally in the machine so that the load is axial.
- Apply the load at a uniform rate without shock.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 05

Efflorescence test on bricks

Aim

To find out the presence of soluble salts in the given bricks by the efflorescence test.

Apparatus / Equipment Required

Brick samples (5 nos.), shallow dish, distilled water, oven, drying rack.

Theory

Efflorescence is the deposition of white, soluble salts on the surface of bricks and masonry when water evaporates. The salts are usually sulphates of sodium, potassium and magnesium present in the brick earth or in the mortar, and their presence indicates poor durability.

In the test, bricks are partially immersed in distilled water and dried alternately for seven days; the deposit is graded as nil, slight, moderate, heavy or serious as per IS: 3495 (Part 3).

Procedure

1. Five bricks were dried in an oven and placed vertically in a shallow dish.
2. Distilled water was poured into the dish up to a depth of about 25 mm.
3. The dish was kept in a well-ventilated room at 20–30 °C for seven days, the water being replenished whenever it evaporated.
4. After seven days the bricks were taken out and dried in an oven at 110 °C.
5. The surface of each brick was examined for white deposits and the efflorescence was graded.

Observation

Brick	Observation after 7 days	Grade of efflorescence
1	No perceptible deposit	Nil
2	Very small deposit (less than 10% area)	Slight
3	No perceptible deposit	Nil
4	Small deposit covering about 10% area	Slight
5	No perceptible deposit	Nil

Water used: distilled water. Grading as per IS: 3495 (Part 3).

Calculation

Nil — no perceptible deposit.

Slight — deposit covers not more than 10% of the surface.

Moderate — deposit covers about 50% of the surface.

Heavy — deposit covers more than 50% of the surface, with powdering and flaking.

Serious — heavy deposit with powdering and flaking of the surface.

Result

Out of the five bricks tested, three showed **nil** efflorescence and two showed **slight** efflorescence. Since the deposit is only slight, the bricks are **acceptable** for masonry work; bricks showing moderate or heavy efflorescence should be rejected.

Precautions

- Use distilled water, not tap water, to avoid adding salts.
- Maintain the water level throughout the seven-day period.
- Dry the bricks in the shade before the final oven drying.
- Examine the bricks in good light and compare with the standard plate.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 06

Field tests and fineness of cement

Aim

To carry out the field tests on a cement sample and to determine its fineness by sieving.

Apparatus / Equipment Required

Cement sample, 90 µm IS sieve, weighing balance (0.1 g), airtight container; Glass of water, trowel, non-porous plate.

Theory

Cement is a finely ground binding material which, when mixed with water, sets and hardens. Field tests such as feel, colour, presence of lumps and the float test give a quick idea of the quality of cement at the site, while the fineness test gives an indication of the rate of strength gain.

Fresher and finer cement gives better strength, but cement that has absorbed moisture from the air forms lumps and loses strength; such cement should not be used for important work.

Formulae Used

- Percentage residue = (mass of residue / mass of sample) × 100

Procedure

1. The colour, feel and presence of lumps in the cement sample were observed.
2. A small quantity of cement was thrown into a glass of water and its floating behaviour was noted.
3. A handful of cement was rubbed between the palms to check for grittiness.
4. A 100 g sample was sieved on a 90 µm IS sieve for 15 minutes and the residue was weighed.
5. The percentage residue was calculated and the quality of the cement was judged.

Observation

Test	Observation	Inference
Colour	Uniform grey with greenish tinge	Fresh OPC
Feel	Smooth, not gritty	Fineness satisfactory
Lumps	No hard lumps	No moisture absorption
Float test	Floats for a short time before sinking	Good quality cement
Mass of sample / residue	100 g / 8.4 g	Residue 8.4%

Cement: OPC 43 grade, fresh stock. Observations are sample values.

Calculation

Percentage residue = (8.4 / 100) × 100 = **8.4 %**

A residue of less than 10% on the 90 µm sieve indicates cement of satisfactory fineness.

Result

The cement sample passes all the field tests — uniform colour, smooth feel, no lumps and the float test — and its residue on the 90 µm sieve is **8.4%**, which is within the permissible limit. Hence the cement is of **good quality and satisfactory fineness**.

Precautions

- Collect the cement sample from a freshly opened bag.
- Store the sample in an airtight container to prevent moisture absorption.
- Clean the sieve before and after sieving.
- Do not judge the fineness by touch alone; confirm it by sieving.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 07

Determination of initial and final setting time of cement

Aim

To determine the initial and final setting time of the given cement sample using the Vicat apparatus.

Apparatus / Equipment Required

Vicat apparatus with plunger and needle, balance, measuring cylinder, mixing tray, trowel; Stopwatch, glass plate, cement sample (400 g).

Theory

Setting is the stiffening of the cement paste after mixing with water. The initial setting time is the period up to which the paste remains workable, and the final setting time is the period after which the paste has completely hardened.

As per IS: 4031 (Part 5), the initial setting time of ordinary Portland cement shall not be less than 30 minutes and the final setting time shall not exceed 600 minutes. These times govern the placing and finishing of concrete and mortar.

Formulae Used

- Initial setting time = time of 5 ± 0.5 mm penetration – time of adding water
- Final setting time = time at which the ring fails to make an impression – time of adding water

Procedure

1. A cement paste of standard consistency was prepared (400 g cement with 30% water).
2. The paste was filled in the Vicat mould and placed on the base plate.
3. The needle was lowered at intervals of 5 minutes and the penetration was observed.
4. The time when the needle penetrated only 5 mm from the bottom was recorded as the initial setting time.
5. Testing was continued with the annular ring needle until it failed to make an impression on the surface of the paste.
6. The elapsed times were calculated and compared with the IS limits.

Observation

Time after adding water	Penetration / impression	Inference
15 min	Full penetration	Paste plastic
35 min	Penetration 12 mm	Setting started
50 min	Penetration 5 mm	Initial setting time
5 h 30 min	Ring makes impression	—
7 h 00 min	Ring fails to impress	Final setting time

Water added = 30% of mass of cement (standard consistency). Readings are sample values.

Calculation

Initial setting time = **50 minutes** (greater than the minimum of 30 minutes ✓)

Final setting time = **7 hours = 420 minutes** (less than the maximum of 600 minutes ✓)

Both values satisfy the requirements of IS: 4031 (Part 5).

Result

The initial setting time of the given cement is **50 minutes** and the final setting time is **420 minutes**. Both are within the limits specified by IS, hence the cement is suitable for mortar and concrete work.

Precautions

- Start the stopwatch exactly at the moment water is added.
- Use the correct needle (1 mm square) for setting time and the annular ring for final setting.
- Keep the mould in a place free from vibration and draught.
- Maintain a room temperature of 27 ± 2 °C.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 08

Determination of compressive strength of cement mortar cubes

Aim

To determine the compressive strength of the given cement by testing 70.6 mm cement mortar cubes prepared with standard sand.

Apparatus / Equipment Required

70.6 mm cube moulds, vibrating machine, standard sand, mixing tray, trowel; Compression testing machine, curing tank, weighing balance.

Theory

The compressive strength of cement is determined on mortar cubes of 1 part cement and 3 parts of standard sand, with water equal to (P/4 + 3)% of the total mass, where P is the standard consistency. The test gives an indication of the grade of the cement.

The cubes are cured in water at 27 ± 2 °C and tested at 7 and 28 days. The 28-day strength of OPC 43 grade cement shall not be less than 43 N/mm².

Formulae Used

- Water to be added = (P/4 + 3)% of (cement + sand) mass
- Compressive strength = failure load / area of cube (70.6 × 70.6 = 4984 mm²)

Procedure

1. 200 g cement and 600 g standard sand were mixed dry and water was added as per the formula.
2. The mortar was mixed for 3 minutes and filled in the cube moulds in three layers.
3. The moulds were vibrated for 2 minutes and kept covered for 24 hours.
4. The cubes were demoulded and cured in water until the date of testing.
5. Three cubes were tested at 7 days and three at 28 days in the compression testing machine.
6. The compressive strength of each cube and the mean strength were calculated.

Observation

Age	Cube No.	Failure load (kN)	Area (mm²)	Strength (N/mm²)
7 days	1	196	4984	39.3
7 days	2	201	4984	40.3
7 days	3	198	4984	39.7
28 days	1	258	4984	51.8
28 days	2	264	4984	53.0
28 days	3	261	4984	52.4

Cement OPC 43 grade; P = 30%, so water = (30/4 + 3) = 10.5% of 800 g = 84 ml. Readings are sample values.

Calculation

Water required = (30/4 + 3)% of 800 g = 10.5% of 800 = **84 ml**

Mean 7-day strength = (39.3 + 40.3 + 39.7)/3 = **39.8 N/mm²**

Mean 28-day strength = (51.8 + 53.0 + 52.4)/3 = **52.4 N/mm²**

As 52.4 N/mm² > 43 N/mm², the cement satisfies the requirement of OPC 43 grade.

Result

The compressive strength of the given cement is **39.8 N/mm² at 7 days** and **52.4 N/mm² at 28 days**. The 28-day strength is more than the specified 43 N/mm², hence the cement conforms to **OPC 43 grade**.

Precautions

- Use standard sand and weigh all materials accurately.
- Vibrate the moulds for exactly 2 minutes for all cubes.
- Cure the cubes in water at 27 ± 2 °C and test them on the due date.
- Keep the loading faces of the cube parallel while testing.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 09

Determination of flow (workability) of cement mortar by flow table

Aim

To determine the flow value of a cement mortar mix of given proportion using the flow table.

Apparatus / Equipment Required

Flow table with brass cone and 25 mm drop arrangement; Mixing tray, trowel, weighing balance, tamping rod, measuring cylinder, cement and sand.

Theory

Flow of mortar is a measure of its consistency and workability. In the flow table test the mortar is placed in a standard cone, the table is dropped 25 times through a height of 25 mm, and the increase in the average diameter of the mortar is expressed as the flow value.

The flow value depends on the water-cement ratio and the proportion of sand. A mortar for masonry work generally has a flow of about 105–115%, while plaster requires a slightly higher flow.

Formulae Used

- Flow (%) = [(average diameter after drops − base diameter) / base diameter] × 100
- Average diameter = (d■ + d■ + d■ + d■) / 4

Procedure

1. Cement and sand in the proportion 1:4 by mass were mixed dry, and water was added gradually to get a workable mix.
2. The flow table was cleaned and the brass cone was placed at its centre.
3. The mortar was filled in the cone in two layers, each tamped 20 times, and the top was struck off level.
4. The cone was lifted vertically and the table was dropped 25 times in about 15 seconds.
5. Four diameters of the spread mortar were measured and their average was calculated.
6. The flow value was computed as a percentage of the base diameter (250 mm).

Observation

Trial	d■ (mm)	d■ (mm)	d■ (mm)	d■ (mm)	Mean dia. (mm)	Flow (%)
1	272	276	274	270	273	109.2
2	270	274	272	268	271	108.4
3	274	278	276	272	275	110.0
Mean	—	—	—	—	273	109.2

Mortar 1:4 by mass, water-cement ratio 0.55. Base diameter of table top = 250 mm. Readings are sample values.

Calculation

Mean diameter = 273 mm

Flow = (273 − 250)/250 × 100 = **9.2 %** (i.e. flow value = 109.2%)

Flow value = 100 + 9.2 = **109.2 %**, which is within the usual range of 105–115% for masonry mortar.

Result

The flow value of the given 1:4 cement mortar at a water-cement ratio of 0.55 is **109.2 %**. The mortar has satisfactory workability for masonry and plastering work.

Precautions

- The flow table must be level and firmly fixed.
- Lift the cone vertically without twisting or shaking.
- Give exactly 25 drops at a uniform rate of about one drop per 0.6 second.
- Measure the diameters along four equally spaced directions.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 10

Study of timber as a building material — identification, seasoning and defects

Aim

To study the properties of timber used in construction and to identify its common defects, seasoning methods and uses.

Apparatus / Equipment Required

Timber samples (teak, sal, deodar, pine, shisham), magnifying glass, pen knife; Moisture meter, weighing balance, steel scale.

Theory

Timber is obtained from trees and is used for doors, windows, roofs, floors, shuttering and furniture because of its light weight, strength and good thermal insulation. Its properties depend on the species, age of the tree, rate of growth and moisture content.

Green timber contains a large amount of sap which causes shrinkage, warping and decay; it is therefore seasoned by air drying, kiln drying or chemical treatment before use. Defects such as knots, shakes, twists and decay reduce its strength and durability.

Procedure

1. The colour, grain, texture and smell of each timber sample were observed.
2. The hardness was tested with a pen knife and the weight was compared by lifting the samples.
3. The moisture content of each sample was measured with a moisture meter.
4. The samples were examined with a magnifying glass for knots, shakes, cracks and other defects.
5. The species were identified and their common uses in building construction were recorded.

Observation

Timber	Colour / grain	Hardness	Moisture (%)	Common use
Teak	Golden brown, straight grain	Hard	10.5	Doors, windows, furniture
Sal	Dark brown, coarse grain	Very hard	11.0	Beams, sleepers, flooring
Deodar	Light brown, fine grain	Soft	9.8	Roof trusses, shuttering
Shisham	Brown, interlocked grain	Hard	10.2	Furniture, doors
Pine	Light yellow, straight grain	Soft	12.0	Ceiling, panelling

Moisture content measured with an electrical moisture meter; readings are sample values.

Calculation

Seasoning reduces the moisture content from about 60–70% in green timber to 10–12% for use.

Percentage shrinkage in volume = (volume green − volume dry)/volume green × 100. For the test sample: (100 − 92)/100 × 100 = **8 %**.

Timber with moisture content above 15% should not be used for joinery work.

Result

Five species of timber were studied. Teak and sal are hard, durable and suitable for doors, windows and structural members; deodar and pine are soft and light and are used for trusses, shuttering and panelling. The moisture content of the seasoned samples is 9.8–12%, which is within the acceptable range.

Precautions

- Store timber on a level surface with spacers so that air circulates freely.
- Protect timber from rain and direct sun during storage.
- Reject samples showing knots, shakes or fungal growth for structural use.
- Measure moisture content at several places on the sample and take the average.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 11

Comparative study of modern building materials — fly ash bricks, AAC blocks and ceramic tiles

Aim

To compare the properties of modern building materials such as fly ash bricks, autoclaved aerated concrete (AAC) blocks and ceramic tiles with conventional burnt clay bricks.

Apparatus / Equipment Required

Samples of fly ash brick, AAC block, burnt clay brick and ceramic tile; Weighing balance, steel scale, water tank, compression testing machine (if available).

Theory

Modern building materials are being adopted to reduce the dead load of structures, to use industrial waste and to improve thermal insulation. Fly ash bricks are made from fly ash, lime, gypsum and sand; AAC blocks are lightweight, aerated, autoclaved cement products; ceramic tiles are used for flooring and wall finishes.

The selection of a material depends on its density, compressive strength, water absorption, thermal conductivity, fire resistance and cost.

Procedure

1. The dimensions and mass of each sample were measured and the density was calculated.
2. Each sample was immersed in water for 24 hours and the water absorption was determined.
3. The compressive strength of the samples was found from the values supplied by the manufacturer, or by testing in the compression testing machine.
4. The surface finish, edges, thermal insulation and fire resistance were observed and compared.
5. The observations were tabulated and the suitability of each material for different uses was noted.

Observation

Property	Fly ash brick	AAC block	Clay brick	Ceramic tile
Size (mm)	230×110×75	600×200×100	230×110×70	300×300×8
Density (kg/m³)	1700	650	1800	2300
Compressive strength (N/mm²)	10.0	4.0	8.4	—
Water absorption (%)	12	20	12	3
Thermal conductivity (W/mK)	0.55	0.16	0.60	1.10
Main use	Masonry	Partition walls	Masonry	Flooring / wall finish

Values are typical manufacturer/standard data; actual laboratory values may vary.

Calculation

Density = mass / volume. For the AAC block: 0.6 × 0.2 × 0.1 = 0.012 m³; mass 7.8 kg → 7.8/0.012 = **650 kg/m³**.

For the clay brick: volume = 0.23 × 0.11 × 0.07 = 0.001771 m³; mass 3.19 kg → 3.19/0.001771 = **1800 kg/m³**.

Saving in dead load with AAC blocks = (1800 – 650)/1800 × 100 = **64 %**, which reduces the structural cost of the building.

Result

Fly ash bricks have good strength (10 N/mm²) and use an industrial waste; AAC blocks are much lighter (650 kg/m³) and give better thermal insulation but lower strength; ceramic tiles have very low water absorption and are best suited for flooring and wall finishes.

Precautions

- Handle AAC blocks carefully as they are brittle and easily damaged at the edges.
- Soak fly ash bricks before use only if specified by the manufacturer.
- Use tiles of the specified grade for flooring and skirting.
- Check the manufacturer's test certificate for strength and density values.

Date: _____ Teacher's Signature: _____

PRACTICAL NO. 12

Preparation of labelled sketches of building components — doors, windows, stairs and roofs

Aim

To prepare labelled sketches of the common building components — doors, windows, staircases and roofs — and to study their parts and functions.

Apparatus / Equipment Required

Drawing sheet (A3), drawing board, T-square, set squares, scale, pencil, eraser, compass.

Theory

Doors and windows are openings in walls provided for access, light and ventilation. A door consists of a frame, shutters, panels, and fittings such as hinges, handles and locks; windows may be fixed, sliding or pivoted.

Stairs provide vertical circulation within a building and consist of steps (riser and tread), landing, stringer and handrail. Roofs are classified as pitched or flat and consist of the covering, purlins, rafters and trusses.

Procedure

1. The drawing sheet was fixed on the board and a border and title block were drawn.
2. A plan and elevation of a panelled door (1.0 m × 2.1 m) were drawn to a scale of 1:20.
3. A plan and elevation of a sliding window (1.2 m × 1.2 m) were drawn next to the door.
4. A section of a dog-legged staircase was drawn showing riser, tread, landing and handrail.
5. Sketches of a pitched roof (king post truss) and a flat roof were drawn with labels.
6. All parts were labelled neatly and the sheet was completed with a title block.

Observation

Component	Parts	Standard dimensions / proportion
Door	Frame, shutter, panel, hinges, handle, lock	Width 0.9–1.2 m, height 2.0–2.1 m
Window	Frame, shutter, glass pane, sill, lintel	Sill height 0.9 m, area 1/8 of floor area
Staircase	Riser, tread, landing, stringer, handrail	Riser 150–175 mm, tread 250–300 mm
Pitched roof	Ridge, eaves, purlin, rafter, truss, covering	Slope 1:3 to 1:5
Flat roof	Slab, waterproofing, parapet, drain	Slope 1:50 for drainage

Sketches drawn to scale 1:20 for doors/windows and 1:50 for stair and roof.

Calculation

Stair calculation: for a floor height of 3.0 m and riser 165 mm, number of risers = 3000/165 = **18.2** ≈ **18**, hence 17 treads (number of treads = risers − 1).

Width of staircase = 1.0 m per flight for a residential building.

Ventilation check: window area required = 1/8 of floor area = 12/8 = **1.5 m²** for a room of 12 m²; the window of 1.2 m × 1.2 m provides 1.44 m², which is slightly less and should be increased.

Result

Labelled sketches of doors, windows, a dog-legged staircase and pitched and flat roofs were prepared to a suitable scale. The standard dimensions and proportions of each component were studied; for a floor height of 3.0 m, 18 risers of 165 mm and 17 treads of 280 mm are required.

Precautions

- Draw all lines with a sharp pencil and to the stated scale.
- Keep the proportions correct — a wrong proportion makes the drawing misleading.
- Label each part clearly with leader lines.
- Do not fold the drawing sheet; submit it flat or rolled.

Date: _____ Teacher's Signature: _____

References and Useful Standards

Practical file likhne ke liye nimnlikhit textbooks, IS codes aur curriculum outcomes ka sahayata liya gaya hai. Viva mein IS code number poochhe jaate hain, isliye inhe ek baar zaroor padh lein.

S. No.	Reference / Standard
1	Sushil Kumar, *Building Construction*, Standard Publishers Distributors.
2	B.C. Punmia, *Building Construction*, Laxmi Publications.
3	M.S. Shetty, *Concrete Technology* (for cement and mortar tests), S. Chand.
4	IS: 1077, IS: 3495 (Parts 1–3) — bricks and their tests.
5	IS: 4031 (Parts 1–6) — tests on cement; IS: 2250 and IS: 5512 — masonry mortar and flow table.
6	IS: 2185, IS: 12894, IS: 15622 — concrete blocks, fly ash bricks and AAC blocks.
7	IS: 287 and IS: 399 — timber; SP: 7 National Building Code; IS: 962 — staircases.
8	SBTE Bihar curriculum for Building Construction & Material (2415304 / P2415304).

Note: Is file mein di gayi observation readings sample hain — samajhne aur format ke liye. Apne practical class mein actual readings le kar observation table dobara bharein aur result calculate karein. Practical ki sankhya apne college ke lab manual ke anusaar ho sakti hai.